

Science TOPICS

FOR SENIOR STUDENTS

PETER GRIBBEN AND
WENDY ST GERMAIN



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Science TOPICS

FOR SENIOR STUDENTS

BOOK

1

**PETER GRIBBEN AND
WENDY ST GERMAIN**



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The three *Science for Senior Students* books present a wide variety of material aimed at students studying science subjects in Years 11 and 12. Many of the work sheets can be used as extension work for more able Year 9 and 10 students. The activities are designed to gain the interest of students and extend their knowledge, understanding and recall of science, something not always encouraged by conventional text books.

Over the three books in this series many branches of science are represented, including astronomy, biology, chemistry, ecology, geology and physics.

The broad range of subject areas is covered in a variety of styles to suit most ability levels. These include comprehension exercises, assignments, tests, graphing exercises, and calculations. The majority of topics are covered in a single page, with none being more than two pages.

The design highlights definitions, and key concepts, separate from the main text.

Many of the sheets can be used by teachers who have little prior experience in science.

In order to extend self-motivated students, and improve their fact-finding skills, most topics include a 'research' section. Outline answers to these are provided in the Answers section.

The worksheets in *Science for Senior Students* are intended to:

- provide relevant material for class use to cover the absence of the regular teacher
- introduce new work in an easily assimilated manner
- provide homework or extension exercises
- review and revise specific topics prior to tests/examinations
- cover the 'gaps' left by standard text books.



Where calculations or graphing work are included on a work sheet, these are indicated by the use of an icon graphic.



Contents

A FOUNDATIONS OF CHEMISTRY

1 & 2	An Elemental Tale: Assignment	5
3 & 4	Matter	7
5 & 6	Chemical Reactions	9
7 & 8	Separation Techniques	11

B FORCES AND MOTION

1	Newton's Laws of Motion: Law I	13
2	Newton's Laws of Motion: Law II	14
3	Newton's Laws of Motion: Law III	15
4	Simple Machines I	16
5	Simple Machines II	17
6	Simple Machines III	18
7	Motion: Using equations I	19
8	Motion: Using equations II	20

C ELECTRICITY AND MAGNETISM

1	Electricity I	21
2	Static Electricity	22
3	Resistance I	23
4	Resistance II	24
5	Output of a Power Supply	25
6	Fuses and Current	26
7	Transformers	27
8	Electrical Terms	28

D WAVES

1	Sound	29
2	Ultrasound	30
3	Hearing Loss	31
4	Noise and Health	32
5	Sound: Extended questions	33
6	Radiation	34
7 & 8	Looking at Light: Cloze	35

E GEOLOGY

1 & 2	Inside the Earth	37
3	The Rock Cycle	39
4	Tectonic Plates: Assignment	40
5	Folds and Faults	41
6	Geological Periods	42
7	Rocks and Minerals: Questions	43
8	Rocks and Minerals: Cloze	44

Answers		45
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[illegible]Science Topics for Senior Students Book 1

An Elemental Tale: Assignment

'Okay Auntie, if (Eu) _____ the horse, I'll be on my way.' Having finished his refreshment, he mounted his (B) _____ but reliable old horse, Neddy, who had been eating his Aunt's (Ge) _____ s outside. He had his shooting (Fe) _____ strapped to his side. 'Gotta go, I've got important (Bi) _____ in town. Come on Neddy!' He dug his spurs into the old horse and (Rh) _____ off into the distance.

He headed out for the bright (Ne) _____ lights of Gribbville, aiming to rob the stagecoach. He had heard there was a load of precious (U) _____ being sent from the diggings at the old (Br) _____ which belonged to his elder brother, whom he hated. He promised that if he ever saw his kinsman again he would shoot him and then (Ba) _____. If, however, the stagecoach didn't turn up, he had a (Ra) _____ on the town bank planned.

He didn't like the bank manager or the people in the saloon. He had (Pd) _____ at cards and lost a lot of money. When he realised how much cash he had wasted over the years he nearly threw a (Ta) _____.

Soon he spotted the stage, guarded by the Sheriff, who wore a (Sn) _____ badge. 'Halt', yelled the outlaw, 'or I'll fill you full of (Pb) _____'. The lawman drew his gun in response, but was too slow. The Kid's gun blazed like burning (Mg) _____, it seemed the life of the (Cu) _____ wasn't worth a plugged (Ni) _____. The only passenger inside the coach, a (Pt) _____ blonde, feared for her life.

Suddenly, a shout rang out, 'Hi Ho

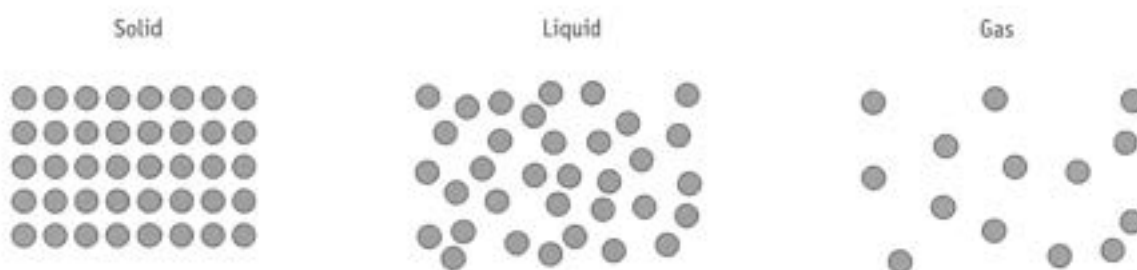
(Ag) _____ !' and a masked man could be (Xe) _____ a white horse racing across the (Si) _____ sands.

A (H) _____ bomb could not have stopped the disguised avenger. The career of the outlaw was over — doomed to spend his life behind (Cr) _____ steel bars. A (C) _____ copy of what happened to all who went against the Lone Stranger. Many have tried to beat him, but they (Ar) _____, all in jail.

Matter

Substances can exist as solids, liquids or gases. All matter is made up of tiny particles (small pieces) called atoms. How these atoms are arranged in a substance determines its state.

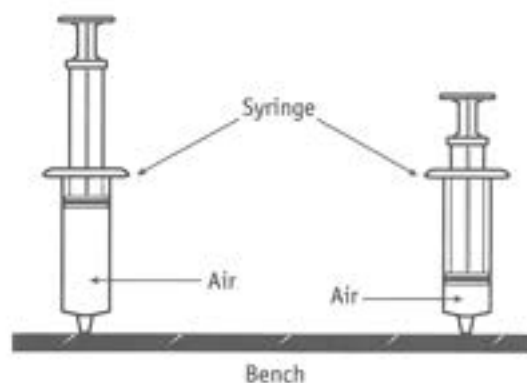
Matter: any physical substance or material. It is anything that takes up space.



- 1 Describe the arrangements of particles in solids, liquids and gases.

Lauren was doing experiments on the volumes of solids, liquids and gases. In one experiment she filled a syringe with air. The ends were sealed so no gas could go in or out. She then pushed in the plunger as hard as she could.

- 2 Draw two diagrams showing how air particles are arranged. The first should show how the particles are before they are compressed and the second should show them after they have been compressed.
- 3 Do you think a solid could be compressed to occupy a smaller volume? Explain your answer.
 - Particles in a solid are held together by strong forces and can only vibrate to and fro.
 - Particles in a liquid can move independently.
 - Particles in a gas are only loosely held together and move very quickly in all directions.
- 4 Draw a diagram to show how particles move in a solid.
- 5 Draw a diagram to show how particles move in a liquid.
- 6 Draw a diagram to show how particles move in a gas.



Lauren's partner, Jess, then took a clean, dry syringe and filled it with granules of instant coffee.

- 7 Would Jess be able to squash the granules into a smaller volume? Give a reason for your answer.

In another group, Demelza, Deborah and Peter were investigating the properties of modelling clay. They had been told by their teacher to put some water into a measuring cylinder and make a note of the volume in millilitres (mL). They then put a piece of modelling clay in the measuring cylinder and measured how much the water level rose. The volume of the modelling clay was equal to the rise in water level.

- 8 Draw a labelled diagram of what you think the measuring cylinder used to investigate the modelling clay would look like before the clay was added.
- 9 Draw a diagram of the measuring cylinder after the clay was added.

Matter

The pupils then tried another experiment. They made the same piece of clay into many different shapes and then put them in the water contained in the measuring cylinder to measure the volume.

- 10** Do you think that the volume of the modelling clay would change when it was a different shape? Explain your answer.

When a solid is heated, energy is put in and the particles vibrate more rapidly. The solid expands and, if enough energy is put in, the particles can move apart from each other and form a liquid.

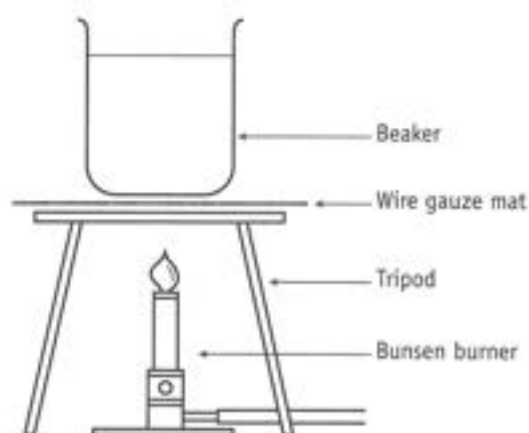
- 11** What do we call this change from a solid to a liquid?

If a liquid is heated strongly enough, the particles can have sufficient kinetic (moving) energy to break away from each other to form a gas.

- 12** What do we call this change from a liquid to a gas?

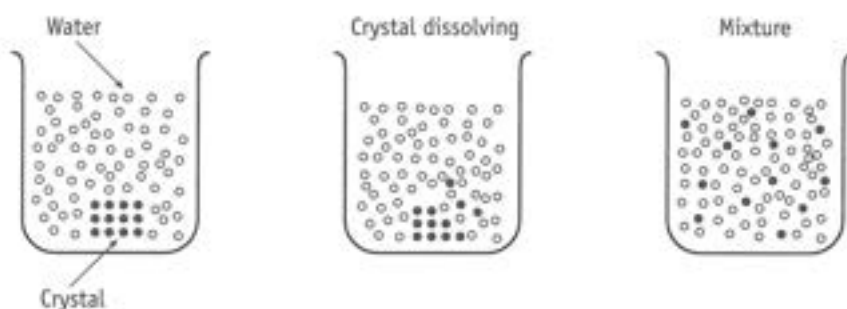
If you leave some water in an open beaker, the liquid level will slowly drop and eventually all of the water will disappear.

- 13** What do we call it when water dries up in this way?



DIFFUSION

The diagrams below show a simple example of diffusion. They represent what happens when a crystal, of copper sulfate for example, dissolves in water.



- 14** Use the particle theory of matter to explain what is happening in the diagrams above.

Diffusion: is the movement of particles from an area of higher concentration to an area of lower concentration. It stops when particles are evenly distributed.

- 15** Use the concept of diffusion to explain why you can smell perfume from a bottle a distance away a short while after the bottle is opened.
- 16** What does 'kinetic' mean?
- 17** Define 'matter'.



RESEARCH

- 18** What is 'sublimation'?
- 19** What is 'condensing'?
- 20** What is 'Brownian motion'?

Chemical Reactions

WHAT IS A CHEMICAL REACTION?

Changes occur around us all the time. Bread goes dry, milk goes sour, steel nails rust, seeds sprout, water evaporates and so on. We can classify these changes as either *physical* or *chemical*.

In a physical change, some of the properties of the substance change, but the substance is still the same. For example, when liquid water is frozen and it changes to ice, its properties change. It is now a solid and cannot be poured — but it is still water. It still has the formula H_2O (that is it contains two hydrogen atoms and one oxygen atom per molecule) and can easily be changed back to liquid water by warming it.

- 1 What is the chemical symbol for hydrogen?
- 2 What is the chemical symbol for oxygen?

Ordinary light globes have thin coils of wire made of the metal tungsten (W) inside them. These are called *filaments*. When electricity passes through a filament it becomes white hot and gives out light (and heat). Even though it looks different, the filament is still tungsten. This is a physical change because *no new substance has been formed*. When you turn the light off, the filament is the same as it was before you switched the light on.



A *physical* change can usually be *reversed*. In a chemical change new substances are formed. Chemical changes are called *chemical reactions* and invariably involve energy changes.

Burning is a chemical reaction. In a bushfire, vegetation is burnt, producing smoke and invisible gases, and leaving ash (mainly carbon) behind. The smoke, gases and ash are new substances — they are completely different from the original vegetation. The reaction also produces heat and light.

- 3 What substance must be present before something can burn?

Another (slower) chemical reaction is the corrosion (rusting) of iron. Iron (Fe) is a shiny silvery metal and it can combine with oxygen (O), a clear colourless gas, to produce Fe_2O_3 .

- 4 What is the name of the compound with the formula Fe_2O_3 ?

This new, brown substance has properties different from the original iron and oxygen. It is difficult to get the iron back without using a lot of energy. One form of iron oxide is haematite, Fe_2O_3 .

- 5 What is the ratio of iron to oxygen atoms in haematite?

Copy the following passage choosing the correct response from the words provided:

A change of state, for example melting, is a (6) *physical/chemical* change. In a physical change a new substance (7) *is/is not* formed. A chemical (8) *reaction/substance* causes new products to be formed, and energy changes are (9) *always/never* involved.

Most *chemical* changes *cannot* easily be reversed. In a chemical reaction we divide the substances into two groups — *reactants* and *products*. The reactants are the materials you start with and which react with each other. The products are the new substances produced in the reaction.

Chemical reactions only occur under certain conditions. Often heat is needed to cause a chemical reaction. You need heat to fry an egg in a pan. Sometimes you need electricity to cause a chemical reaction. This is what happens when you charge a car battery. Some reactions are slow, such as corrosion (rusting). Others occur very quickly, such as explosions.

Chemical Reactions

During a *chemical* reaction an *energy change* also occurs. All life depends on chemical reactions. Green plants use a chemical reaction called *photosynthesis* to make food (glucose, $C_6H_{12}O_6$) from carbon dioxide (CO_2), water and the energy from sunlight.

Chemical reactions occur in our bodies to digest the food we eat. Tooth decay is caused by chemical reactions between our teeth and acids in food. To make dough rise when you are making bread, you need to put it in a warm place.

Sometimes people want to slow down or stop some chemical reactions. Combustion (burning) is not always a welcome reaction. People are put in danger when an uncontrolled fire occurs at home, for example, if a chip pan catches fire. Burning needs three things: heat, fuel and oxygen. These can be shown in a 'Fire triangle'.



10 Copy the diagram of a fire triangle and complete it by labelling in the three things needed for fire.

11 What would happen if we remove any one of the things mentioned in your answer to Question 10?

If you want to stop food spoiling you put it in a refrigerator or freezer. The lower the temperature, the slower the reaction occurs. We can also stop iron (and steel) objects from rusting by coating them in paint. This stops oxygen from reaching and reacting with the metal.

Reactions which give out heat energy are called *exothermic* reactions. Reactions which take in heat energy are *endothermic* reactions.

12 Give one main difference between chemical changes and physical changes.

13 Divide the following into two groups, physical changes and chemical changes: making toast, water boiling, a candle burning, freezing an 'icy pole', cooking a sausage, an apple rotting, clothes drying on a clothes line and gas burning in a bunsen burner.

14 Give the chemical symbols for iron and oxygen.

15 Give the name and symbol for the metal used to make the filaments for electric light globes.

16 What is meant by the chemical formula for water being expressed as H_2O ?

17 Give one example of a very fast reaction.

18 Give one example of a very slow reaction.

19 Give another name for 'corrosion'.

20 Explain what is meant by 'reactants' and by 'products'

Some reactions are useful and we want them to happen. Other reactions are not wanted and cause us problems.

21 Give one example of a wanted reaction.

22 Give one example of a unwanted reaction.

23 What substance has the formula $C_6H_{12}O_6$?

24 Is a candle burning an exothermic reaction or an endothermic reaction? Explain your answer.



RESEARCH

25 Suggests some ways you could prevent a metal from corroding.

Separation Techniques

Pure substances contain only one type of particle. Elements such as hydrogen (H), oxygen (O), and copper (Cu) are each only made up of one type of atom. The purity of gold (Au) is shown by karats (carats). Twenty-four karat gold is pure gold.

- 1 A gold ring is 18 karat gold. What percentage of it is gold?

Compounds such as carbon dioxide (CO_2) and common salt (sodium chloride, NaCl) can also be pure substances if only molecules of these substances are present. Water (H_2O) is a compound. Water can be purified by distillation — boiling and condensation of vapour. Distilled water is a pure substance because it contains *only* water. Tap water is a mixture — it contains dissolved solids.

- 2 Are the contents of a bottle labelled 'pure mineral water', really pure? Explain your answer.

The air around us is a mixture — it is composed mainly of nitrogen (N, approximately 80%) and oxygen (approximately 20%). These elements are mixed, not combined.

- 3 What is another name for the air around us?

- 4 Separate the substance listed below into mixtures and pure substances. (Put your answer in the form of a table.)

sea water, distilled water, soda water, water, copper sulfate, 18 karat gold, air, lead, silver, oxygen, copper sulfate solution, blood, milk, carbon dioxide, petrol, soil, helium, iron, steel, sugar

DISSOLVING, FILTERING AND EVAPORATING

Two solids can be separated if one is soluble in a particular solvent (such as water or ethanol) and the other is not. A sand and salt mixture can be separated by this technique:

- add the mixture to the water, heat and stir (the soluble salt will dissolve in the water and the insoluble sand won't)
- filter the mixture (the salt solution will pass through the porous filter paper and the sand will not)
- recover the salt by evaporation.

- 5 What is meant when we call filter paper 'porous'?

- 6 Could you separate a mixture of sucrose (sugar) and sodium chloride, using the method outlined above, with water as a solvent? Explain your answer.

DISTILLATION

The liquid in a mixture of a liquid and solid can be recovered by distillation. The mixture is heated, the liquid turns to a vapour. This vapour passes through the inner part of a Liebig condenser where it is cooled and turns to a liquid again. The liquid that has been distilled is the distillate.

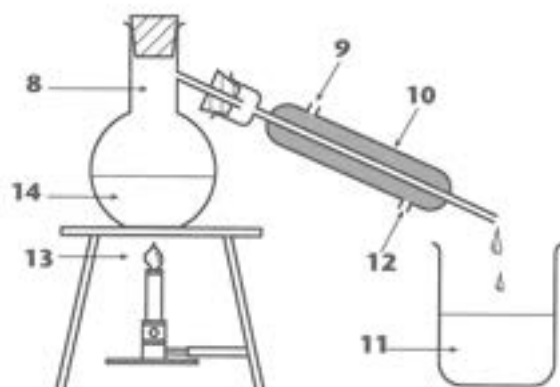
- 7 Copy the diagram on the right.

Label the numbered parts of the diagram, using the following terms:

cold water in distillate heat Liebig condenser solution vapour water out

If two or more miscible liquids have different boiling points, then they can be distilled by a technique called fractional distillation. An ethanol ('metho') and water mixture is an example of this. Ethanol has a boiling point of 78°C and will distill off (by becoming a vapour) before water which has a boiling point of 100°C .

- 15 What is meant when two liquids are 'miscible'?



Separation Techniques

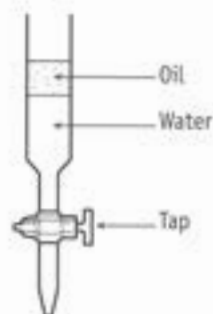
SEPARATING (SEPARATORY) FUNNEL

Liquids that do not mix with each other, that are *immiscible*, can be separated using a separating funnel, as shown.

- 16 Oil and water are immiscible. Explain how a mixture of oil and water could be separated using this apparatus.

Density is defined as mass divided by volume, and is expressed as g mL^{-1} .

- 17 Which is less dense, oil or water? Explain your answer.



SUBLIMATION

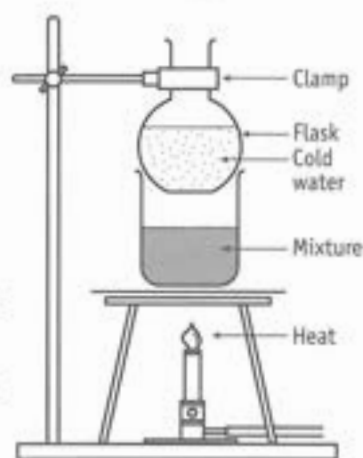
When a solid changes directly to a gas (and vice versa) without changing to a liquid, it is said to *sublime*. Not many substances display this property. Carbon dioxide (CO_2), ammonium chloride (NH_4Cl) and iodine (I) are examples.

$\text{CO}_2(\text{s}) \rightleftharpoons \text{CO}_2(\text{g})$, represents the sublimation of carbon dioxide.

- 18 Write another equation showing the sublimation of ammonium chloride.

The apparatus shown on the right can be used to separate a substance that sublimes from one that doesn't.

- 19 Explain how this apparatus could be used to separate ammonium chloride from sodium chloride.

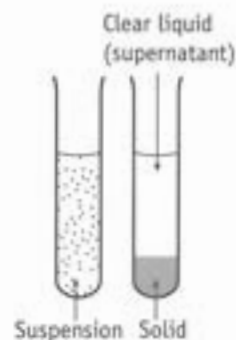


CENTRIFUGING

This technique can be used to separate small amounts of suspended solids from a liquid. A centrifuge spins test tubes very quickly, sometimes many thousands of revolutions per minute (rpm). This causes the particles in the suspension to be flung to the bottom, leaving an upper layer of clear liquid — sometimes called the *supernatant*.

The liquid can be poured out (decanted) leaving the solid behind. Centrifuges have many applications — one being to separate red blood cells from blood. In blood the supernatant is plasma (mainly water) and the solid is mainly blood cells.

- 20 Name a household appliance that works in a similar manner to a centrifuge.
21 Could a centrifuge be used to remove salt from sea water? Give a reason.

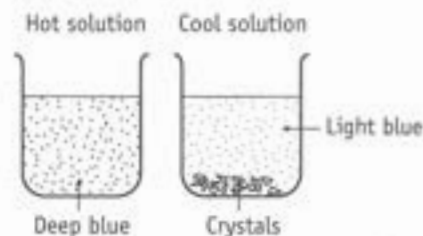


CRYSTALLISATION

Some soluble substances form crystals. These solids are usually more soluble at high temperatures than at low temperatures. Because of this, many solids can be separated from solutions by allowing them form crystals. Copper (II) sulfate is one example.

If a hot, saturated solution of copper sulfate is allowed to cool, then crystals will begin to appear. These crystals can be removed by filtration.

- 22 Apart from crystals appearing, what other change would you note in the cooler solution?
23 How would you turn a dilute solution of copper sulfate to a more concentrated one without adding any more copper sulfate.
24 Explain what is meant by 'decant'.
25 Explain what is meant by 'supernatant'.
26 Explain what is meant by 'saturated solution'.
27 Explain what 'sublimation' is?



Newton's Laws of Motion: Law I

Isaac Newton was a brilliant scientist and mathematician. As well as his work on mechanics, he made many important contributions to astronomy and optics (the study of the nature of light). Mechanics is a branch of physics which looks at objects and how they behave when forces act on them. Newton's three laws of motion relate force and motion.

Newton's first law: If an object is at rest, or has a constant speed in one direction, then there is no net (resultant/unbalanced) force acting on it.

- 1 Find another definition of Newton's first law. Copy it and compare it to the version above.

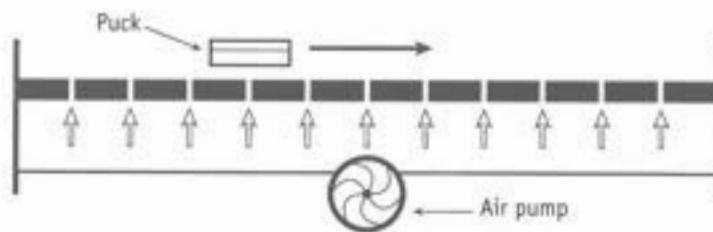
Another way of looking at this is to say that, if there is no net force acting on an object, its acceleration is zero. That is, it will not change its motion (or lack of it). This doesn't necessarily mean that a stationary object has no forces acting on it, just that these forces are balanced. Think about a TV set sitting on a table.



- 2 What force is pulling the TV set downwards?
- 3 What must the table be doing to stop the TV set from falling downwards?
- 4 Consider a spacecraft travelling well away from the effect of the gravitational pull of any stars or planets. What will happen if the rocket engine used to propel the craft suddenly stops? Explain your answer.

When two surfaces are moving over each other then the force of *friction* acts to resist motion.

Air hockey is a popular game. An electric blower forces air through a lot of small holes in the playing surface. The puck (a small disc) floats on a film of air.



If the puck is hit, it will travel for a long distance before it stops or slows down.

- 5 Explain why the puck can travel so far on an air hockey table.
- 6 Galileo stated, in the late 1500s, that an object resists changes in its motion. Does this agree with Newton's first law? Explain your answer.
- 7 An unbalanced force can cause an object to change speed or direction. True or false?
- 8 Explain why it is dangerous to have loose objects in the back of a moving car.



RESEARCH

- 9 Define 'dynamics'.
- 10 Find out where and when Newton did his research.
- 11 Name two contributions made by Galileo in his studies of motion.
- 12 Name two contributions made by Newton to the study of astronomy.



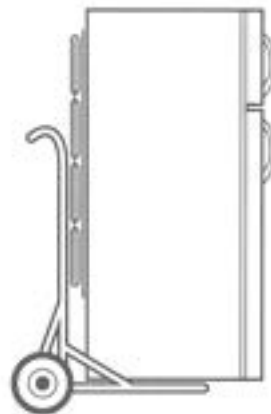
Newton's Laws of Motion: Law II

Dynamics is the study of the relationship between the motion of an object and the forces acting on it. Isaac Newton (1642–1727) was a scientist who lived in England and gave us many of the laws used in dynamics. Newton's first law explains what happens to an object that has no net (unbalanced) forces acting on it. The object will either remain at rest or continue to move in a straight line at constant speed. Newton's second law deals with what happens to an object that has an unbalanced force acting on it.

Imagine you are pushing an object and friction is negligible. If you push the object with a force, F , then the object will move with acceleration, a . If you apply twice the force, then the acceleration will double.

If we apply force F to an object twice the mass, then acceleration will be $0.5a$.

- 1 What will be the acceleration of an object of half the original mass when a force of $2F$ is applied to it?



Newton's second law: The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to the mass of the object.

$$\text{Force} = \text{mass} \times \text{acceleration}$$

$$F = ma$$

The unit of force is the *newton* (N). One newton is the force needed to accelerate a mass of 1 kg by 1 m/s/s. About 5 N is needed to switch on a light, 20 N is needed to open a drink's can, 2000 N is the force applied to a hard-hit tennis ball and over 200 000 N can come from a jet engine.

- 2 Is a force of 1 N large or small? Explain your answer.

A cyclist accelerates by pushing harder on the pedals. Imagine that the cyclist and bicycle have a total mass of 80 kg and the acceleration is 0.5 m/s/s. How can we work out the force involved?

$$\text{Force} = \text{mass} \times \text{acceleration} = 80 \text{ kg} \times 0.5 \text{ m/s/s} = 40 \text{ N}$$

- 3 A 2000 kg car is accelerating at 3 m/s/s. What force is required for this? Show your working.
- 4 There are two cars from the same manufacturer, with identical engines providing the same amount of power. Car 'A' has a mass of 2500 kg, and car 'B' has a mass of 1800 kg. Which car will be capable of the greater acceleration? Explain your answer.

Cartoons often show 'superheroes' who can accelerate to immense speed, without using force. Just by lifting an arm, they can zoom into the sky.

- 5 Do you think that Newton would have believed in these superheroes? Explain your answer.

INERTIA

Objects tend to resist a change in velocity, that is they resist a force trying to speed them up or slow them down. This property is called *inertia* and is measured as mass. The greater the mass of an object, the greater its inertia.

- 6 Which will have the greater inertia, a 1 kg bag of sugar or a 1 kg block of lead? Explain your answer.

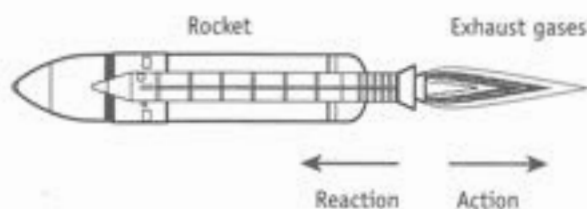
Newton's Laws of Motion:

Law III

You may have heard the phrase used, when talking about a simple matter, 'It doesn't take a rocket scientist to ...' This implies that rockets are complicated objects. The technology may be complex, but the principle behind rockets is simple. It is based on Newton's third law of motion which tells us that single, isolated forces cannot exist.

Newton's third law: Forces occur in equal and opposite pairs called the action and reaction. For every action, there is an equal and opposite reaction.

In a rocket motor, a fuel burns with an internal supply of oxygen and produces high velocity exhaust gases. The rocket is pushed in the opposite direction. A jet engine works on a similar principle but uses air from the atmosphere.



1 Why can't jet engines be used in space?

Another way of stating Newton's third law is that a force that changes the motion of an object must come from another object. This (second) object has itself an equal and opposite force exerted on it by the first object.

2 Explain in terms of Newton's third law, what happens between a hammer and the nail when you hammer a nail into a piece of wood. Include the terms 'action' and 'reaction' in your answer.



Newton's third law also has applications in sports. Imagine a soccer player kicking a ball. The player's moving foot exerts a force on the ball, accelerating it in the direction of the movement of the foot. The ball exerts an equal and opposite force on the foot, which is felt by a sudden slowing down of the foot.

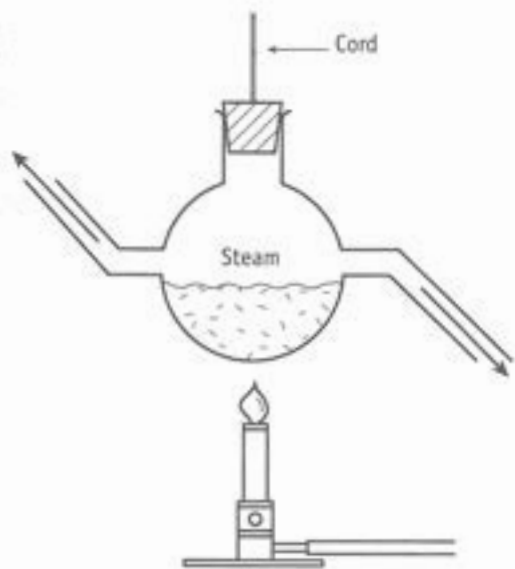
3 Explain, in similar terms, what happens when a tennis player hits an oncoming ball with a racquet.

4 What happens to the shape of a tennis ball at the moment it is struck with a racquet?

5 A fisherman stepped from his boat onto a jetty. He neglected to tie the boat securely, so the boat moved away from the jetty as soon as he stepped from it. Use a simple, labelled diagram and Newton's third law to explain this incident.

6 The diagram shows a simplified version of 'Hero's engine', invented over 2000 years ago. This device consists of a sealed flask containing water suspended by a cord. When the water is heated it boils and turns to steam which comes out of two tubes pointed in opposite directions. Explain what happens to the flask.

7 It is possible to have a single isolated force. True or false?



RESEARCH

8 Find out when and where Hero's engine was invented.

Simple Machines I

Energy changes things. It is the ability to do work and make things happen. A force is something that produces or changes movement, for example, a push, pull, twist or turn. Forces can also change the shape of an object. Forces are measured in newtons (N). Pushes and pulls are called *contact* forces. *Non-contact* forces are gravity, magnetic forces, electric forces and nuclear forces.

Friction is a force which opposes motion. It occurs when two surfaces slide or move over each other. Rough surfaces create more friction than even surfaces. Friction can be lessened in machines and other places by using smooth surfaces and lubricants.

- 1 What is meant by a 'lubricant'?

Friction is not always a nuisance.

- 2 Explain why tyres on cars wouldn't work without friction.

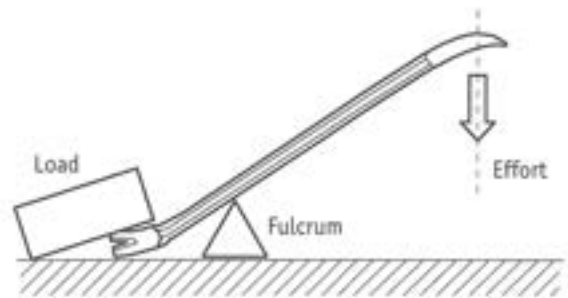
A *machine* is an object that changes force or motion (movement). Machines transfer energy from place to place, and make work easier. They do not do less work nor use less energy.



Machine: a device which is used to overcome a force called the 'load' by applying another force called the 'effort'.

Machines can be force multipliers (increasing the force applied to the load), speed multipliers (increasing the distance moved by the load) and they can also change the direction of movement of a force.

Levers are simple machines. A lever is a bar turning about a fixed point. Every lever must have an effort, a fulcrum (pivot) and a load. The effort is where we apply a force. The fulcrum is the point about which the lever moves. The load is where the force acts. The diagram shows a crow bar, which is a lever.



- 3 Name two contact forces.
- 4 Name two non-contact forces.
- 5 What unit is used to measure forces?
- 6 What three things are present in all levers.
- 7 Name two ways in which friction can be reduced.
- 8 'Machines do less work than if the job was done by hand.' True or false?
- 9 What can forces do to an object as well as change its speed?



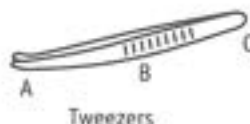
RESEARCH

- 10 A scientist called Archimedes is credited with defining the principle of the lever. Find out when and where Archimedes lived.

Simple Machines II

Below are shown a series of levers (not drawn to scale). The effort, load and fulcrum (pivot) are labelled A, B and C — but not necessarily in that order.

- 1 For the pliers, identify the effort, load and fulcrum.
- 2 For the tweezers, identify the effort, load and fulcrum.
- 3 For the wheelbarrow, identify the effort, load and fulcrum.



- 4 Explain how the pliers could make work easier in a specific application.
- 5 Explain how the tweezers could make work easier in a specific application.
- 6 Explain how the wheelbarrow could make work easier in a specific application.

Effort $\times$ distance (from the fulcrum) = load $\times$ distance (from the fulcrum).



- 7 What effort would be needed to lift a load of 25 N using the lever system shown at left? Show your working.
- 8 What is another name for a 'fulcrum'?

PULLEYS

A pulley is a wheel with a groove to carry a rope. The rope is pulled at one end (the effort) to lift a weight (the load) at the other end. In a single pulley the distance moved by the effort should be the same as the distance moved by the load. The effort should be the same as the load (plus any friction).

9 Single pulleys just change the direction of a force. True or false?
If more than one pulley is used, for example a block and tackle, then the effort should be less than the load. In the system shown in the diagram (using two pulleys) the effort would move twice the distance of the load. Therefore the effort should be half that of the load (plus any friction).

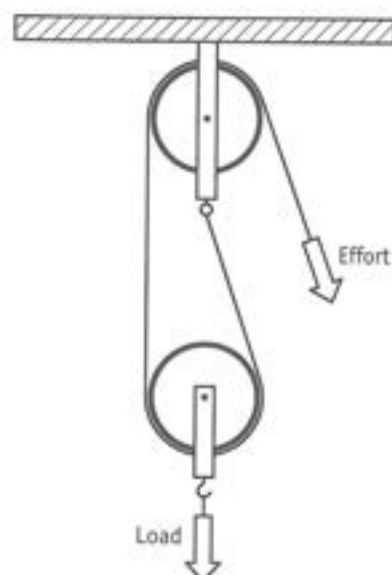
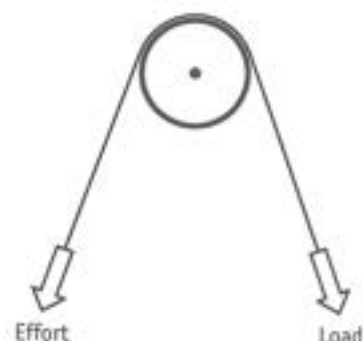
10 If a pulley system had three pulleys how far would the effort move compared to the load?

11 If a pulley system had three pulleys how would the effort needed compare to the load?

12 Name one use for a pulley system.

RESEARCH

13 Archimedes invented compound pulleys over 2000 years ago. He is also famous for the 'Archimedes screw'. Find out how this device works.





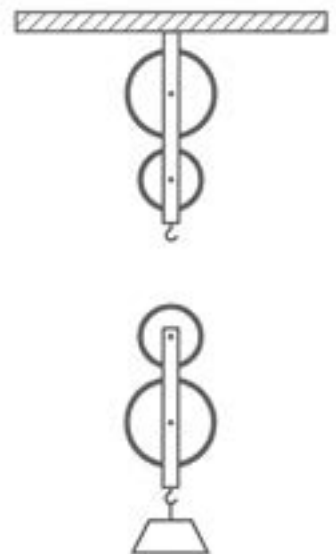
Simple Machines III

The *velocity ratio* of a machine is the distance moved by the effort compared to the distance moved by the load, that is, distance moved by effort $\div$ distance moved by load. For example, if we are using a lever (such as a crowbar) where the effort moves 40 cm and the load moves 10 cm, then the velocity ratio is $40 \div 10 = 4 : 1$.

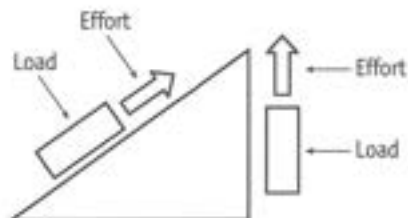
The *mechanical advantage* is the load compared to the effort, that is load $\div$ effort. Theoretically the mechanical advantage should equal the velocity ratio, but it is less because no machine is 100% efficient (that is not all effort put in is turned into useful work). For example, if we put in 25 kg effort into a machine (with a push or a pull) and the machine moves a 100 kg load, then the mechanical advantage is (load $\div$ effort) $100 \div 25 = 4 : 1$

- 1 Explain why a machine, such as a compound pulley, is unlikely to be 100% efficient.

The diagram on the right shows an incomplete pulley system. In a compound pulley, the velocity ratio equals the number of pulleys.



- 2 Copy the diagram of the pulley system.
- 3 On the diagram, show how a rope or cord would be connected to the pulley wheels.
- 4 Label the system on the diagram showing load and effort.
- 5 What is the velocity ratio of this machine?
- 6 Assuming that the machine shown is 100% efficient, what effort would be needed to lift a load of 600 kg?

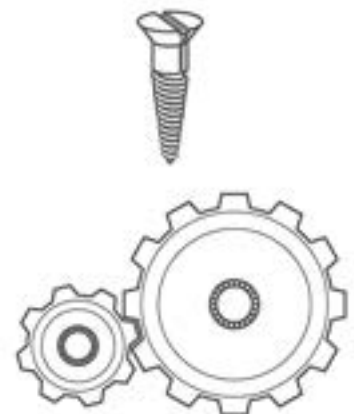


The *inclined plane* is a slanting surface, or slope. It is one of the oldest machines known to humans. It takes less force to pull a mass up an inclined plane than it does to lift it vertically. Ramps are sometimes used rather than steps. Although less effort may be used, the work is done over a larger distance. Inclined planes include screws, nuts and bolts. Vices and car jacks are examples of screws.

- 7 Explain why a screw is an example of an inclined plane.

Wheels were invented between 3500 BC to 3000 BC. The earliest wheels were solid wooden discs rotating on a round axle. The invention of the wheel made life easier and helped in the advance of human civilisation. Wheels can be changed to do different tasks.

- 8 Give three different examples of wheels being used to make work more easy.
- 9 Wheels can be fitted with teeth. What do we call wheels with teeth?



RESEARCH

- 10 Some historians believe that ancient civilisations may have used ramps to erect large stone structures. Explain how they could have done this. (You may want to study Stonehenge, the megaliths on Easter Island or Egyptian obelisks.)



Motion: Using equations 1

$$\text{Speed} = \text{distance} \div \text{time}$$

$$\text{Distance} = \text{speed} \times \text{time}$$

$$\text{Time} = \text{distance} \div \text{speed}$$

For each problem below identify the equation to use. Substitute the given values and solve the problem. Show your working for all problems. Don't forget to include the units used in your answer. The first one has been done for you.

- 1 A rifle bullet travels 1500 metres in 2 seconds. What is its average speed?
Speed = distance $\div$ time = $1500 \text{ m} \div 2 \text{ s} = 750 \text{ metres per second (m. s}^{-1}\text{)}$
- 2 A long distance runner travels 20 kilometres (km) in 100 minutes. What is her average speed?
- 3 A car takes 8 hours to travel the 650 km from Sydney to Albury. What is the average speed for the trip?
- 4 A swimmer is moving at 2 kilometres per hour. How long will it take, at that speed, to swim 24 kilometres?
- 5 A missile travels at 2000 kilometres per hour. How far will it travel in 1.5 hours?
- 6 A passenger jet takes off from Melbourne and flies non-stop for 12 hours before re-fuelling. If its average speed is 500 kilometres per hour, how far will it have travelled in the 12 hours?
- 7 A bowler was measured bowling a ball at 100 kilometres per hour. If the ball travels 20 m from the bowler's hand to the bat, how long does this take?

$$v = u + at$$

The final velocity (v) of an object is equal to its initial velocity (u) added to acceleration (a) multiplied by time (t). A deceleration (slowing down) is shown by a negative value for a .

- 8 What is the final velocity of a cyclist initially travelling at 5 m. s^{-1} who accelerates at 2 metres per second per second (m. s^{-2}) for 3 seconds?
- 9 A skateboarder is travelling at 10 m. s^{-1} and must make an emergency stop, coming to a halt in two seconds. What is the acceleration of the skateboarder?
- 10 A speedboat is travelling at a steady speed and then accelerates for 8 seconds at 3 m. s^{-1} . The boat's final speed is 50 m. s^{-1} . What was its initial speed?
- 11 A light truck making deliveries is travelling at 8 m. s^{-1} and then accelerates at 2 m. s^{-2} for a total of 5 seconds. What is the final speed of the truck?
- 12 Express your answer to Question 11 in kilometres per hour.
- 13 A bobsleigh starts from rest and accelerates uniformly (steadily) for 8 seconds when its speed reaches 30 m. s^{-1} . What is the acceleration of the vehicle during this period?



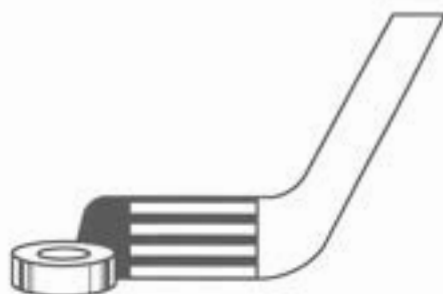
Motion: Using equations II

For each problem decide which equation to use. Substitute the given values and solve the problem. Show your working for all problems. Don't forget to include the units used in your answer. Assume that friction is negligible. The first one has been done for you.

Force (F) equals mass (m) multiplied by acceleration (a). Force is measured in newtons (N), mass is measured in kilograms (kg) and acceleration is measured in metres per second per second (m. s^{-2})

$$F = m \times a$$

- 1 An ice puck of mass 0.25 kg is accelerating at 20 m. s^{-2} . What is the force on the puck?
Force (N) = $0.25 \text{ kg} \times 20 \text{ m. s}^{-2} = 5 \text{ N}$
- 2 A soccer player strikes a stationary ball of mass 500 g with a force of 100 N. What is the initial acceleration of the ball?
- 3 A golfer hits a 45 g ball with a force of 3000 N. What is the initial acceleration of the ball?
- 4 When a particular object is pushed with a force of 200 N it accelerates at 10 m s^{-2} . What is the mass of the object?
- 5 Gravity is the force which attracts objects to the centre of the Earth. It causes an acceleration of 9.8 m. s^{-2} . What is the force of gravity on a person of mass 70 kg?



The distance (s) travelled by an object is equal to the initial (starting) velocity (u) multiplied by time (t) plus half of the acceleration (a) multiplied by the square of time (t).

$$s = ut + \frac{1}{2}at^2$$

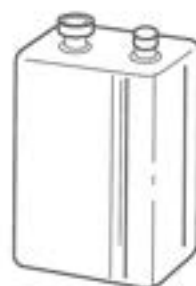
- 6 A car starts from rest and accelerates at a steady 4.0 m. s^{-2} for a total of 8 seconds. How far does the car travel in this time?
Distance = 0 (starts from rest) + $0.5 \times 4.0 \times 8^2 = 128 \text{ m.}$
- 7 A motorcycle starts from rest and accelerates uniformly for 400 m on a race track. It took 15 seconds to cover this distance. What was the vehicle's acceleration?
- 8 In a particular car race the vehicles had a 'rolling start'. This means that they travel at a steady speed and can only accelerate at a given position or signal. One vehicle was travelling at 20 m. s^{-1} when the signal was given. Five seconds later the car was travelling at 30 m. s^{-1} . What distance had the car covered in this time?
- 9 A plane lands at a speed of 80 m. s^{-1} . It takes 20 seconds to come to a halt on a 1000 m runway. What is its rate of acceleration?
- 10 A vehicle started from rest and was accelerated at 5 m. s^{-2} steadily until it had covered 100 m. How long did it take to cover this distance?

Electricity I

Electricity is used to provide a versatile source of energy. It makes our lights work and, without it, we wouldn't have radio, television, computers and CD players. We take electrical light globes (bulbs) for granted, but they were only invented after a lot of hard work. Thomas Alva Edison was an American inventor who patented, in 1880, an electric light filament similar to those used in light globes today. Sir Joseph Swan (an English scientist) came up with the idea for the incandescent electric lamp at the same time.

Electricity is the movement of electrons (negative particles) along a conductor. This flow of electrons is called an *electric current*. For electricity to flow, it must have a complete pathway or circuit to follow. If there is a gap, the circuit is broken and no current flows.

Energy can be stored in cells (sometimes called batteries) for conversion to electrical energy when needed. Batteries are used when only low voltage (e.g. 3 volts to 12 volts) of electricity is used in calculators, watches and personal stereos for example. Batteries are also used when things need to be portable. The first battery was invented by the Italian Count Alessandro Volta in about 1800. The unit of electrical measurement called the volt (V) was named after him.



Some electrical devices need large amounts of electricity to operate them. They are powered by mains electricity (240 volts). Using batteries to supply large amounts of electricity is not always practical — they would be too expensive and take up too much space. Today people use a lot of electricity, at home and at work. This electricity is generated at power stations. The first, very small, generator was made by the English scientist Michael Faraday in 1831. Modern power stations are huge and use coal, oil, gas, flowing water, or sometimes nuclear fuels, to provide electricity. In these power stations water or steam drives turbines and generators to produce electrical power.

The voltage leaving a power station can be as high as 350 000 volts. It is carried via high voltage powerlines to local substations (transformers) where it is decreased to 240 volts before it is supplied to homes. Electricity can make our lives easier, but it can also be dangerous. Always be careful when using anything electrical — especially near water.

- 1 Name four things not mentioned on this sheet that use electricity for power.
- 2 What is an 'incandescent electric lamp'?
- 3 Who invented the first battery?
- 4 What type of charge do electrons have?
- 5 What voltage is the mains electricity used in houses?
- 6 What do we call a flow of electrons?
- 7 Who made the first electrical generator?
- 8 What happens at 'substations'?
- 9 Why don't we use batteries to power all electrical devices?
- 10 Name three fuels used by modern power stations.
- 11 Why might it be dangerous to use an electrical hair drier near a bath full of water?



RESEARCH

- 12 Find out one other invention made by Thomas Edison.
- 13 Find out one other invention made by Joseph Swan.

Static Electricity

Electrons are small particles which carry a negative (–) electric charge. If a substance gains electrons it will gain a negative charge.

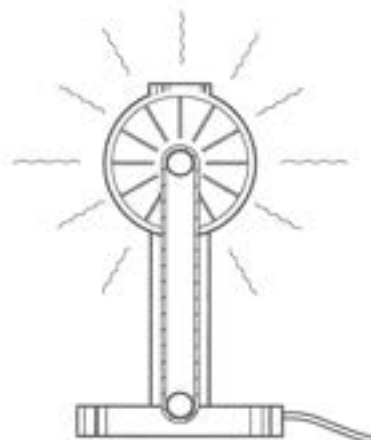
- 1 What will happen to a substance which loses electrons?

When an insulator is rubbed with a cloth, it can become charged with 'static electricity'. Static means stationary — it doesn't flow like mains electricity. If a polythene rod is rubbed with a cotton sheet, the polythene takes electrons away from the fibre and the rod is negatively charged.

- 2 What will be the charge on the cotton cloth?

Static electricity can be discharged if the charged substance is 'earthed' (connected to the ground), which neutralises the charge.

The Van de Graaff generator is a machine for the generation of extremely high voltages, developed in the 1930s by the American physicist, Robert Van de Graaff. It consists of a high-voltage terminal, often an aluminium sphere, mounted at the top of an insulating column. A belt runs from a pulley at the base of the column to a pulley inside the sphere. Inside the sphere a metal comb is mounted parallel to the moving belt and puts electrons onto the belt. The belt carries its charge up into the sphere and passes them on to the sphere.



- 3 Why must the sphere be placed on an insulating column?
- 4 What charge should be carried by the sphere after the machine has been running for a while?
- 5 On a rainy day a science teacher was demonstrating a Van de Graaff generator in the laboratory, but was not having much success. She put the machine away and told the class she would wait until a dry, hot day to do the experiment again. What was the reason for this?

Benjamin Franklin was an American statesman, scientist, and philosopher. Among his inventions were bifocal spectacles and lightning conductors. He is famous for his kite experiment in 1752. Here is an adapted version of a letter Franklin wrote describing his experiment:

'To the top of the upright of the kite's cross, fix a very sharp pointed wire, rising a foot (30 cm) or more above the wood. To the end of the twine (string), next to the hand, is to be tied a silk ribbon, and where the silk and twine join, a key may be fastened. This kite is to be raised when a thunder gust appears to be coming on, and the person who holds the string must stand under some cover, so that the silk ribbon doesn't get wet. Take care that the twine does not touch the frame of the door or window. As soon as any of the thunder clouds come over the kite, the pointed wire will draw the electric fire from them, and the kite, with all the twine, will be electrified, and the loose filaments of the twine will stand out in every way and be attracted by an approaching finger. When the rain has wet the kite and twine, so that it can conduct the electric fire freely, you will find it stream out plentifully from the key on the approach of your knuckle.'

- 6 Draw a labelled diagram of Franklin's experiment.
- 7 What was Franklin investigating?
- 8 Why was a pointed wire put on the top of the kite?
- 9 Why was a silk ribbon put between the key and his hand?
- 10 Why did the loose filaments on the twine stand out?
- 11 Franklin flew his kite *before* the arrival of a full storm. What might happen if this experiment were carried out *in* a lightning storm?

Resistance I

Electricity flows when electrons move through conductors.

- 1 What do we call a substance that doesn't conduct electricity?

Good conductors include the elements copper (Cu), aluminium (Al), silver (Ag) and gold (Au).

- 2 What do these substances have in common?

Electrical resistance: the property of a substance to resist (oppose) the flow of an electrical current

Electrons move easily through good conductors, such as copper which has little resistance to the flow of electrons. They do not move as easily through other substances, which offer more resistance. Not all metals are good conductors like copper. 'Nichrome' wire has a high resistance.

- 3 Nichrome is an alloy of iron (Fe) and two other metals. What are these two metals? (Hint: Ni and Cr.)
- 4 Which circuit will need more energy to push electrons around it, one having low resistance or one having high resistance?

When electricity encounters resistance as it flows in a circuit, it gives off heat. The greater the resistance the greater the heat given off.

- 5 Name three electrical appliances which use electricity to generate heat.
- 6 Why might Nichrome wire, rather than copper, be used in the element of an electrical kettle?

Light bulbs (globes) give out light as well as heat when a current flows through them. The outside of the globe is made of glass containing an unreactive gas such as argon (Ar) or nitrogen (N). Inside there is a coil of tungsten (W) wire called the filament. The filament glows white hot when electricity flows. The unreactive gases stop the wire from burning. If air gets in, the hot filament will react with oxygen in the air and burn.



- 7 Copy the diagram of a light globe, and label the parts indicated by the arrows.
- 8 Does the element tungsten have a high resistance or a low resistance? Give a reason for your answer.



A thin wire has more resistance than a thick one and a long wire has more resistance than a short one.

- 9 Why do you think a thick wire conducts electricity better than a thin one made of the same material?

Copy the following passage and choose the correct words from the pairs written in *italics* to fill in the spaces.

Resistance is the ability of an object to resist the flow of a (10) *current/force*. It is measured in (11) *ohms/amps*. Good conductors have a (12) *high/low* resistance and poor conductors have a (13) *high/low* resistance. Resistance is (14) *directly/inversely* proportional to the cross-sectional area of the resistor and (15) *directly/inversely* proportional to the length.

- 16 List the names and symbols of all the metal elements mentioned on this sheet.



RESEARCH

- 17 What are 'superconductors' and how do they operate?



Resistance II

Ohm's Law: states that the resistance in an electric circuit decides how much current flows for any given voltage in that circuit.

A substance has electrical resistance when it opposes the flow of electrical current. The unit of resistance is the ohm, (named after the German scientist, George Ohm). A resistor has the value of one ohm (Ω) if it allows a current of one amp (A) to flow when a voltage of one volt (V) is applied to it. This can be stated in the form of an equation.

Voltage (volts)	Current (amps)	Resistance (ohms)
12	6	2
	0.1	50
240		30
24	6	
	1	8

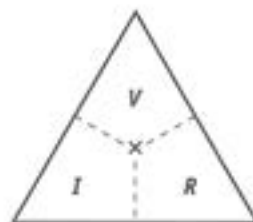
$$\text{Voltage} = \text{current} \times \text{resistance}$$

- 1 Copy the table above and use the given equation to calculate the missing values.

The general symbol for:

- voltage is V (and it is measured in volts, V)
- current is I (and it is measured in amps, A)
- resistance is R (and it is measured in ohms, Ω).

- 2 Use the information above, including the equation, to explain how the triangle at right can be used in calculations involving Ohm's Law.



When resistors are in series (one after the other), they give a resistance of the combined values. If two 10 ohm resistors are in series they give a resistance of 20 ohms.



- 3 Why will these two resistors give a higher resistance than a single resistance of the same value?

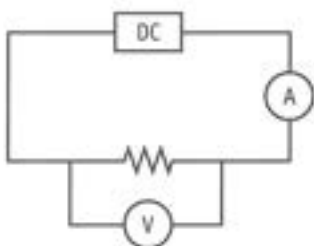
The diagram to the right shows one representation used in circuit diagrams of a resistor.



- 4 Draw a circuit diagram showing a cell, switch and two eight ohm resistors arranged in series.
5 What would be the total resistance of the resistors in the circuit in Question (4)?

When measuring current in a circuit we use an *ammeter* which is wired in series and, when measuring voltage we use a *voltmeter*, wired in parallel. A student set up a circuit as shown below and measured the current flow and voltage drop using several resistors, one at a time. The results are shown below.

- 6 Copy and then label the components of the circuit.



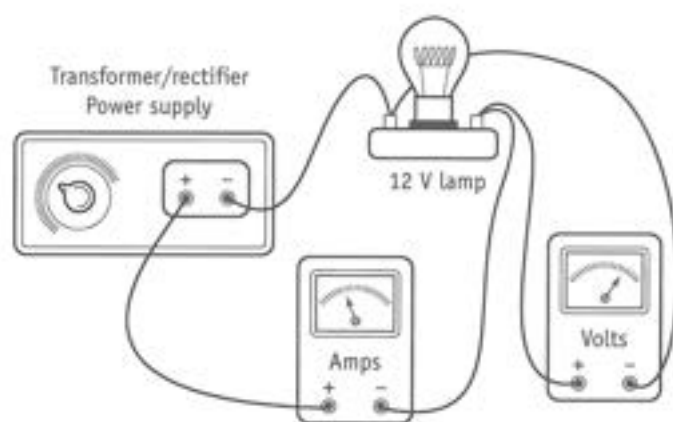
Current (A)	Voltage (V)	Resistance (Ω)
0.5	12	
0.1	6	
0.25	3	
3.3	7	
10	240	

- 7 Copy the table and calculate the value of the individual resistors used.



Output of a Power Supply

Mains electricity is alternating current (AC). This means that the current reverses itself at regular intervals, that is it changes from positive to negative to positive and so on. Direct current (DC) flows in one direction only. A pupil set up the circuit as shown in the diagram to the right. The aim was to calculate the resistance and power for different voltages in a circuit.



- 1 Was the voltage used in the circuit AC or DC? Give a reason for your answer.
- 2 Re-draw the circuit diagram, using recognised symbols for the meters and the lamp.
- 3 Apart from the readings on the meters, what change might the pupil see in the circuit as the voltage from the power supply was varied?
- 4 Which meter is wired in series? (Hint: *series* is wired one after the other.)
- 5 Which meter is wired in parallel? (Hint: *parallel* is wired side by side.)

$$\text{Resistance (ohms)} = \text{voltage (volts)} \div \text{current (amps)}$$

- 6 Write the above equation in symbol form.

$$\text{Power (watts)} = \text{voltage (volts)} \times \text{current (amps)}$$

- 7 Write the above equation in symbol form.
- 8 Below is a sample set of results. Copy the table and calculate the missing values for power and resistance using the two equations.

Setting on power supply	Reading on Voltmeter (V)	Reading on Ammeter (A)	Power (W)	Resistance (Ω)
A	1.4	0.7		
B	2.9	0.9		
C	4.4	1.1		
D	5.9	1.4		
E	7.4	1.6		
F	8.8	1.8		
G	10.3	2.1		

Look at the electrical circuit again.

- 9 Will the light go out if the voltmeter is disconnected. Explain your answer.
- 10 Will the light go out if the ammeter is disconnected. Explain your answer.
- 11 The lamp is broken during the experiment and must be replaced. The choice is between four lamps 6 V, 12 W; 12 V, 12 W; 6 V, 24 W; 12 V, 24 W. Which would you choose? Give a reason.
- 12 What travels through the circuit when a current is flowing?



RESEARCH

- 13 What is meant by a 'rectifier'?



Fuses and Current

A fuse is a safety device that protects an electric circuit from the effect of too much current flowing. It usually contains a thin strip or wire of metal with a high electrical resistance. If the current in the circuit goes past a certain limit, the metal heats up and melts. This breaks the circuit and the electricity stops flowing.

- 1 Give the name and symbol for the units used to measure resistance.

Fuses contain wires in a ceramic or glass holder.

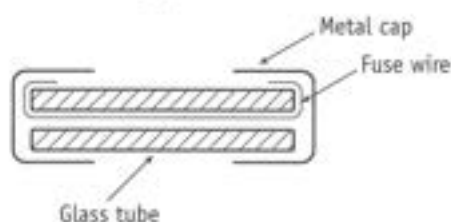
- 2 Why might the fuse wire be contained in a glass tube?

Diagrams A and B on the right show two ways that a fuse can be represented.

Diagram A



Diagram B



- 3 Copy the diagrams.
- 4 Indicate which diagram is likely to be the symbol used in a circuit diagram.
- 5 What type of material is used for the ends (caps) of the type of fuse represented in the second diagram?
- 6 Why is this material used?

$$\text{Power (watts)} = \text{voltage (volts)} \times \text{current (amps)}$$

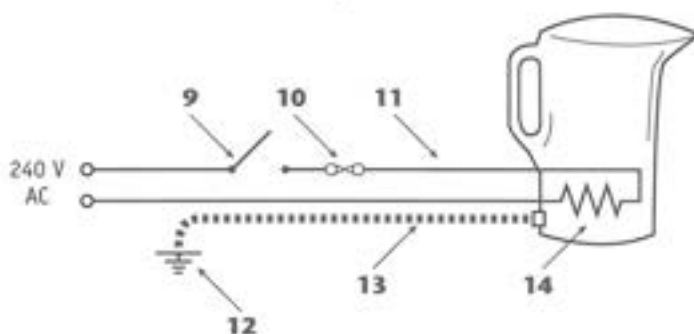
- 7 The power used by an electrical appliance is measured in watts (W). Copy and complete the table below showing the current used by the 240 V electrical equipment. Use the formula shown to calculate the missing values.

Appliance	Watts (W)	Amps (A)
Table lamp	60	
Colour TV	360	
Drill	750	
Hair drier	550	
Heater	2000	

Diagram C shows a simplified circuit diagram of an electric kettle with a metal casing.

- 8 Copy the diagram and label the components (9 to 14)

Diagram C



RESEARCH

- 15 Fuses are being replaced by *circuit-breakers*, which are easier to use and react more quickly. Explain how these work.
- 16 What might happen to house wiring if too much current flows?

Transformers

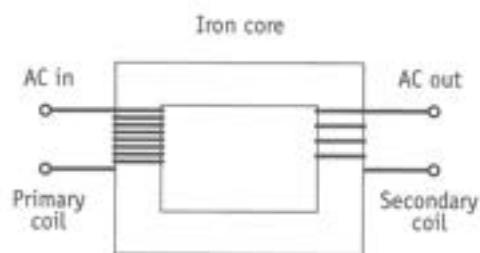
Transformers are devices used to convert AC voltages. Electrical energy coming in is changed and given out at a different voltage by the transformers. They are used in power supplies, audio and television equipment.

1 What is meant by 'AC'?

A transformer consists of a core wrapped in two coils of insulated wire. The primary coil is where the voltage is put in, and the secondary coil is where the (modified) voltage is given out. Household electricity is 240 V AC with a frequency of 50 Hz.

2 What does a frequency of 50 Hz mean for an AC supply?

When the voltage rises and falls in the primary coil it induces (causes) a voltage to flow in the secondary coil. The relationship between voltages and turns of wire in the coils is shown by this equation.



$$\frac{\text{Voltage across primary coil}}{\text{Voltage across secondary coil}} = \frac{\text{turns in primary coil}}{\text{turns in secondary coil}}$$

3 Copy the table below. Use the equation given above to complete the missing values.

Some energy is always 'lost' as heat or noise in transformers, but the transformers can be made as much as 99% efficient. This is done by using special steel alloys for the core. The core is also made of thin insulated sheets ('laminated')

Primary turns	Primary voltage	Secondary turns	Secondary voltage
100	12	1000	120
20	12	100	
50	240		12
	110	12	12
300		2100	1680

4 What does '99% efficient' mean?

5 Why must the wires in a transformer be insulated?

There are 'step-up' and 'step-down' transformers. In a step-up transformer the output voltage is greater than the input voltage.

6 Which is greater in a step-up transformer, the number of turns in the primary coil or in the secondary coil?

7 Is the output voltage in a step-down transformer more than, less than or the same as the input voltage?

In science laboratories, power supply units are often used to change 240 V to 12 V (or less) for use in practical work involving students setting up electrical circuits.

8 Why is this done?

RESEARCH

9 Find out how electricity substations use transformers to change electricity coming from power stations and so on.

10 Give a brief summary (two to three lines) of the discoveries made in electricity by either Nikola Tesla or Hans Oersted.

11 Can transformers change DC voltage? Explain your answer.

Electrical Terms

In the left-hand column of the table there are 26 terms used in electricity. In the right-hand column there are descriptions to go with these terms, but not in the correct order. Match the term with the correct definition/description by writing the correct letters in the left-hand column.

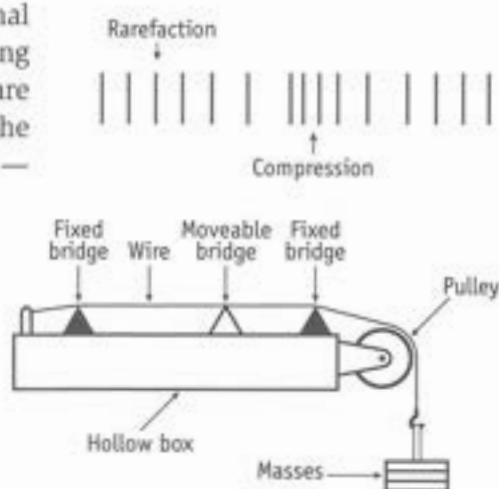


	Answer	Term		Definition
1		electromagnet	a	A small, negative particle.
2		compass	b	A machine that changes electricity to movement.
3		lightning	c	A magnet used to find direction.
4		transformer	d	Type of circuit with 'side-by-side' wiring.
5		power station	e	A substance that doesn't conduct electricity.
6		electron	f	Wire wound around and around.
7		turbines	g	Type of electricity taken to homes by power cables.
8		galvanometer	h	Used to change voltages.
9		cell	i	A device to change electricity to sound.
10		parallel	j	Electricity generated by sunlight.
11		circuit	k	A substance which allows electricity to flow.
12		series	l	Changes electricity to light and heat.
13		insulator	m	These move in power stations as part of generators.
14		globe	n	A sensitive instrument used to measure electricity.
15		hydro electricity	o	Has an electrical supply, wiring and, for example, a globe.
16		loudspeaker	p	Measured in amperes (amps).
17		current	q	A circuit with components wired one after each other.
18		poles	r	Two ends of a magnet.
19		electric motor	s	The type of electricity supplied by batteries.
20		coil	t	A temporary magnet, relying on electricity to work.
21		solar	u	A device using chemicals to produce electricity.
22		direct current	v	Electricity generated by moving water.
23		alternating current	w	A place where electricity is generated on a large scale.
24		conductor	x	A large discharge of static electricity.
25		ohm	y	A type of electricity that doesn't flow.
26		static	z	A unit of resistance.

Sound

Acoustics is the study of sound. Sound waves are longitudinal waves, that is waves which are made up of particles swinging backwards and forwards in the same direction that the waves are travelling. This creates areas of high pressure (compression — the lines close together in the diagram) and low pressure (rarefaction — the lines that are further apart.)

The diagram at right shows a *sonometer* which makes sounds by plucking a wire. The hollow box amplifies the noise made by the vibrating wire. The tension of the wire can be changed by adding or removing masses. The tighter the wire, the higher the frequency or pitch of the sound produced. The length of wire which vibrates can be changed by moving the 'bridge'. The shorter the wire, the higher the frequency.



The same note played on different instruments can sound very different. Although the fundamental frequency may be the same, other vibrations (overtones or harmonics) can also be produced. This distinct sound quality of different instruments is called *timbre*.

Sounds we hear are affected by the movement of the sound, relative to the listener. The 'Doppler Effect' is the change in frequency of a sound when either the listener or the source of the sound is moving. If the object making the noise is moving towards you, then a higher frequency sound is heard. If the object is moving away, then a lower frequency is heard. This is why the note of a fire engine's siren changes as it passes you at high speed.

If you are watching any ball game, you see the ball being kicked or struck and hear the sound a fraction later. This is because light travels much more quickly than sound. The light reaches your eye from the action almost instantaneously, because light travels very quickly — approximately 300 000 kilometres a second. Sound moves much more slowly, at about 330 metres a second (0.33 km per second).

In a similar manner, thunder and lightning occur close together, but you will see the lightning flash a few seconds before you hear the thunder. The sound takes longer than the flash of light to reach you. We can work out roughly how far away lightning is from us. If a lightning flash occurs 3 kilometres away, for example, its light will reach our eyes in $3 \div 300\,000$ seconds, or 0.0001 of a second. The sound would take much longer to reach us, $3 \div 0.33$, approximately 9 seconds. If we measure the number of seconds between the lightning and thunder and divide this by three, we determine the distance to the lightning in kilometres.

- 1 Explain, with the aid of a diagram, why sound is described as a 'longitudinal wave'.
- 2 Explain why a TV viewer might hear the sounds of a football match before someone who is actually inside the ground.
- 3 Give the name of a musical instrument that resembles a sonometer.
- 4 Give two ways in which the note made with a sonometer can be varied.
- 5 What is 'timbre'?
- 6 You are standing on a railway platform when an express train rushes past. It sounds a warning. Explain why this steady note seems to change pitch as the train passes.
- 7 If a flash of lightning occurs 2 km away, how long will it take to hear the thunder? Show your working.



RESEARCH

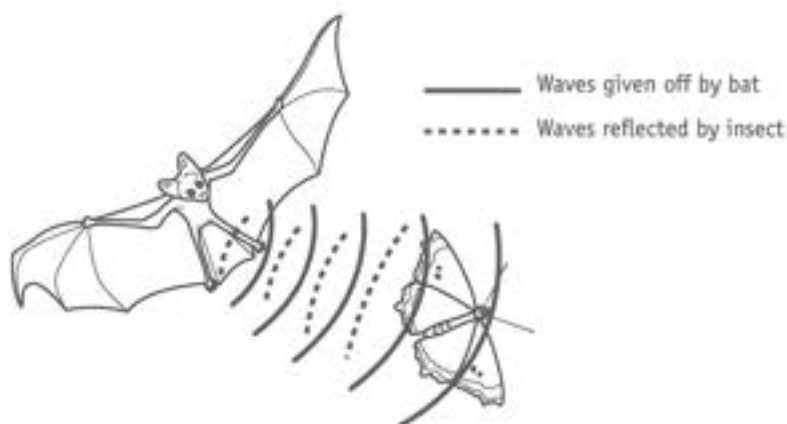
- 8 What is meant when we say jet planes break the 'sound barrier'?
- 9 Explain what is meant by 'resonance' in musical instruments.

Ultrasound

All mammals (including humans) hear sounds using the same types of body parts and in a similar manner. But not all mammals hear the same frequency ranges as humans. 'Silent' whistles, which use frequencies too high for humans to hear, can be used to call dogs. Humans cannot hear sounds much higher than 20 kHz. (1 kHz = 1000 Hz). Frequencies higher than this are called 'ultrasound'. ('Ultra' means 'beyond'.) As we grow older the range of audible frequencies decreases.



An echo is a sound wave which has been reflected off a surface and is heard after the original sound. Ultrasound echoes are often used to find the position of objects. Sonar ('**S**ound **n**avigation and **r**anging') can be used by ships to detect underwater objects. A ship sends out a pulse of sound. The distance from the ship can be calculated using the length of time the echo takes to return to the source.



Bats use ultrasound in a form of echo location. They use this to find prey and avoid objects at night. The bat gives out ultrasonic waves. These are reflected back by an object. The bat can hear ultrasound and knows the position of the object.

Ultrasound can also be used to scan the human body. Muscles, bones and fat all reflect back ultrasound waves differently. The reflected waves can be used (via electronics) to form an

image on a screen. Ultrasound is preferred over X-rays to look at a developing baby in a mother's womb, because it doesn't harm the baby. Ultrasound can also be used to clean jewellery.

- 1 What is the highest frequency that most humans can hear?
- 2 Explain, with the aid of a diagram, how bats find their way around.
- 3 Give three uses that humans make of ultrasound.
- 4 What happens to our hearing as we grow older?
- 5 Why can ultrasound be used to take pictures inside the human body?
- 6 Why might a doctor prefer to take an ultrasound 'picture' of a foetus, rather than an X-ray?
- 7 Why might a keen angler fishing in a well-known lake use a sonar device?
- 8 Why might an oil tanker going through uncharted waters use a sonar device?
- 9 Why might a navy patrol boat practising defence techniques against enemy craft use a sonar device?
- 10 Why might a dog handler use a 'silent whistle' rather than just call the dog or whistle normally?



RESEARCH

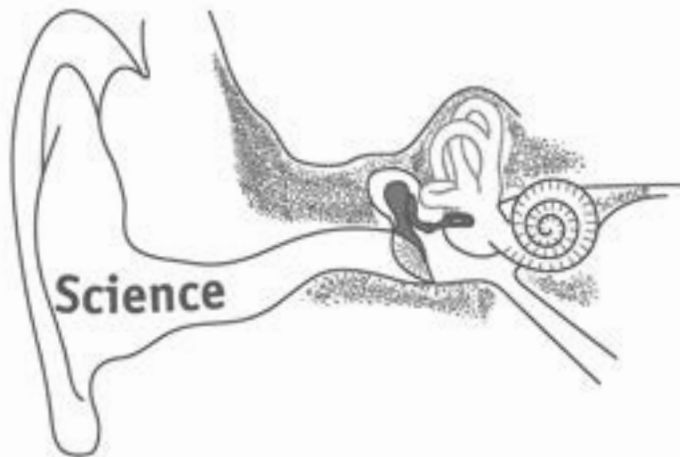
- 11 What is the difference between 'supersonic' and 'ultrasonic'?
- 12 Find out how dolphins use sound to communicate and find their way around.

Hearing Loss



We use units called decibels (dB) to measure the loudness of sound. This is a logarithmic scale, that is, a rise of 10 dB doubles the noise. So, 60 dB is twice as loud as 50 dB and so on.

Type of sound	Sound level (dB)
Jet aircraft taking off	110 to 140
Thunderstorm	90 to 100
High speed train	60 to 90
Talking	50 to 70
Whispering	20 to 50
Falling leaves	10



The average person, aged in the early 20s (provided he or she has not suffered prolonged exposure to loud noise) can hear noises from about 10 dB upwards. As people grow older their hearing becomes less sensitive. The table below shows the quietest sound that can be heard by most people of the ages shown.

Age (years)	25	30	35	40	45	50	55	60	65
Hearing threshold (dB)	10	12.5	15	17.5	20	23	26	30	39

- Use the data in the table to plot a 'best-fit' line graph. Remember to:
 - write your name on the graph.
 - give the graph an appropriate title.
 - plot hearing loss on the y-axis (vertical), showing units used.
 - plot age on the x-axis (horizontal), showing units used.
 - label the axes.
- What would be the quietest noise you would expect an average person aged 62 to hear? Show on your graph how you obtained this answer.
- A person was found to just be able to hear a noise of 14 dB. What would be their approximate age? Show on your graph how you obtained this answer.
- What assumption are you making in your answer to Question 3?
- How much hearing (in dB) has a person 'lost' by the time he or she is 50 years of age?
- How many times louder is a noise of 100 dB compared to one of 90 dB?
- How many times quieter is a noise of 60 dB compared to one of 90 dB?



RESEARCH

- Find out how our lifestyles in Australia can cause hearing loss.
- In developing countries, deafness is often caused by different factors to those in the Western world. Find out what might cause deafness in developing countries.
- Explain how loud noises can damage eardrums.
- Find out what the local regulations are regarding noise levels in a workplace.
- The 'Bionic Ear' is the name used to describe cochlear implants. Describe how cochlear implants can help people with impaired hearing.

Noise and Health

We don't always realise that noise can be a danger to our health. A 'noise' is often defined as 'a loud, disagreeable sound'. Yet the same sound can be either pleasant or a nuisance — it depends on the person hearing it. What you think is 'music' might be 'noise' to older people.

Noise is a problem to people who live in busy cities — they are continually exposed to loud noises. This noise can be from ordinary traffic, airports, trains and trams, building work and machinery. Such types of noise can make life uncomfortable for people living near these sources of noise.

Exposure to loud noise, for example, loud discos and concerts, can damage hearing or even cause deafness. People who have had prolonged exposure to loud noise may suffer from headaches, tiredness and stress.



Noises are measured in units called decibels (dB). Special sound meters are used to measure loudness. Zero decibels is the threshold of audibility, that is, the weakest sound the human ear can hear. Sounds above 140 decibels ('the threshold of pain') hurt your ears and can cause permanent damage. If you whisper, this will give a sound level of about 20 decibels, talking normally will give about 60 decibels, whereas loud rock music registers at about 120 decibels.

We try to keep unwelcome sounds to a minimum. Traffic noise can be a problem so, in the exhaust systems of cars, mufflers (silencers) are included to quieten the noise made by the engine. Insulation can also be used in machinery to reduce its noise level. People working with noisy machines wear ear protectors or earmuffs to reduce the amount of noise entering their ears.

- 1 Define 'noise'.
- 2 What is the unit that we use to measure the loudness of sounds?
- 3 Give four sources of noise in city areas.
- 4 Many rock musicians have become deaf after several years of playing live music. Why is this so?
- 5 Explain what is meant by 'fatigue'?
- 6 Explain what is meant by 'nausea'?
- 7 A petrol lawn mower has a label stuck on the engine, which reads '77 dB'. What does this mean?
- 8 Name two methods you could use to help lessen noise from outside traffic entering your home.
- 9 What is meant by the 'threshold of audibility'?
- 10 What type of protection should people working near aeroplanes at airports use?
- 11 Why do 'personal stereos' carry warnings about volume levels?
- 12 What type of noise most irritates you?
- 13 Imagine this situation. You live in a city that is a popular tourist destination. Many visitors arrive by plane, however the local airport has had a ban on flights arriving or leaving after 11:00 pm or before 6:00 am. The local council wants to extend the hours of operation of the airport. Write a letter that could be sent to a local newspaper using the viewpoint *either* of:
 - a local resident who lives near the airport and is unhappy about the increased noise *or*
 - a local business person who welcomes the possible increased number of visitors.



RESEARCH

- 14 A group of musicians were playing at a local club. They had problems with their sound and said that the club had 'poor acoustics'. What did they mean?



Sound: Extended questions

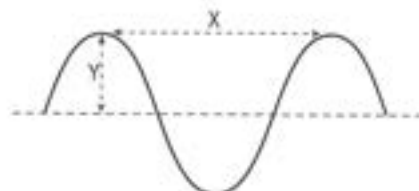
SHOW WORKING IN ALL CALCULATIONS

Ships use echoes to find the depth of water beneath them. A piece of equipment sends a pulse of sound towards the sea floor. The sound bounces back to the ship and the time taken for it to return is measured. Sonar uses this principle.



- In which will it take longer for an echo to return, shallow water or deep water? Explain your answer.
- If the ship is 800 m above the sea bed, how far will a pulse of sound travel, measuring from where it is sent to where it is received by the ship?
- Sound travels at approximately 1500 metres per second in water. What is its speed in air?
Speed, distance and time are linked by the equation shown.
- Use the equation to find how long it will take a pulse to travel from the ship to the sea bed and return, if the ship is in 800 m of water.
- In another situation, it took 1.5 seconds for an echo from the ship to return. How deep was the water?
- Copy the diagram of a sine wave shown on the right.
- On your diagram, label a 'crest'.
- On your diagram, label a 'trough'.
- Explain what is represented by X on the diagram.
- Explain what is represented by Y on the diagram.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$



Copy and complete the following two sentences.

- Sound travels through warm air more _____ than cold air.
- Sound travels faster in liquids and solids than in _____
- What is shown in the diagram on the right?
- Name one type of energy that can be transmitted by the type of wave shown in this diagram.



The relationship between velocity (speed), frequency and wavelength of a wave are shown by the equation on the right.

$$v = f \times \lambda$$

- What is the wavelength (λ) of a sound wave of frequency (f) 5 kHz, travelling at (v) 330 m.s⁻¹?

If an object, making a sound of a steady frequency (pitch), is moving towards you, the sound seems higher in pitch. If the source is moving away it seems lower. You notice this effect when hearing the siren on a fire engine or the whistle of an express train as they go past at speed.

- What is the name of this effect?
- Explain how this effect is generated.
- What unit is used to measure the intensity of sound?

Radiation

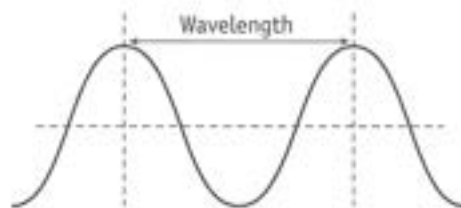
Radiation: any form of energy radiated (sent out) from a source.

Many people, when they hear the word 'radiation', think of nuclear reactions and radioactivity, but the word means much more than that. Three main classes of radiation include electromagnetic radiation (EMR), nuclear radiation, and cosmic radiation. All three types of radiation can travel in a vacuum.

Infra red (IR), ultra-violet (UV), visible (white) light and radio waves are examples of electromagnetic radiation. Radioactivity is a type of nuclear radiation and cosmic radiation (or cosmic rays) come from the sun and other objects in outer space. The sun releases cosmic rays during spectacular eruptions called solar flares.

If you have ever lain on a beach and got a suntan, then the change in skin colour was caused by UV rays from the sun. If you switch on an electric iron, after a while, without touching it, you can feel heat being given off. Heat is a form of IR, which we cannot see. Radio signals are not just transmitted on Earth but occur naturally from outer space, for example from stars, and can be detected on Earth by large radio telescopes.

EMR is always around us and is a form of energy associated with electric and magnetic fields. It behaves like a wave and travels through space at the speed of light, 300 000 kilometres per second. There are different types of electromagnetic radiation. Each type has a different wavelength and frequency, but the same speed. Wavelength is measured in metres. The frequency of a wave is the number of cycles per second passing a certain point. The unit of frequency is Hertz or Hz. One cycle per second is 1 Hz. Different wavelengths of EMR carry different amounts of energy. Short-wavelength radiation carries more energy than long-wavelength radiation.



Large, unstable atoms (*radioisotopes*) can break up and give off energy (gamma rays and X-rays) and particles. Gamma rays and X-rays pass through solid objects and can be used to take pictures of the inside parts of living things. Radioisotopes can be used (with caution) to treat cancers and tumours in humans.

- 1 What type of radiation causes sunburn?
- 2 What is another name for heat radiation?
- 3 Where do cosmic rays come from?
- 4 What is the speed of light in a vacuum?
- 5 What is a 'radioisotope'?
- 6 Name the main types of electromagnetic radiation.
- 7 Explain what is meant by the 'frequency' of a wave.
- 8 Explain, with the aid of a diagram, what is meant by the 'wavelength' of a wave
- 9 Name the units of frequency.
- 10 Name the units of wavelength.
- 11 Why can gamma rays and X-rays help us find out about the human body?
- 12 Which type of electromagnetic radiation carries more energy, short-wavelength or long-wavelength?



RESEARCH

- 13 Briefly describe how radioisotopes can be used to diagnose illnesses.
- 14 Define a 'vacuum'.

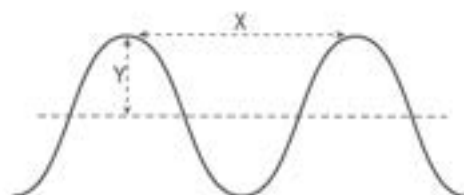
Looking at Light: Cloze

Choose the correct word from the list to fill in the spaces:

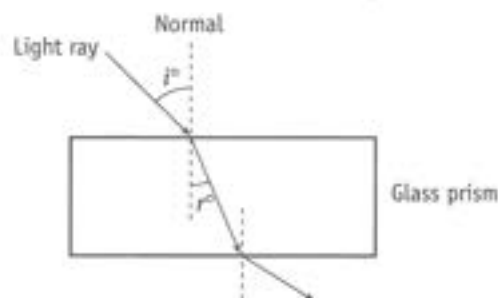
<i>amplitude</i>	<i>apparent</i>	<i>behind</i>	<i>bend</i>	<i>camera</i>	<i>can</i>	<i>concave</i>	<i>converge</i>
<i>convex</i>	<i>cornea</i>	<i>decreases</i>	<i>dense</i>	<i>different</i>	<i>dispersion</i>	<i>diverge</i>	<i>energy</i>
<i>eye</i>	<i>focus</i>	<i>front</i>	<i>glass</i>	<i>incidence</i>	<i>increases</i>	<i>inverted</i>	<i>iris</i>
<i>larger</i>	<i>lens</i>	<i>long</i>	<i>normal</i>	<i>prism</i>	<i>radio</i>	<i>real</i>	<i>reflection</i>
<i>refract</i>	<i>retina</i>	<i>second</i>	<i>seem</i>	<i>short</i>	<i>smaller</i>	<i>spectrum</i>	<i>speed</i>
<i>transparent</i>	<i>transverse</i>	<i>vacuum</i>	<i>water</i>	<i>waves</i>	<i>wavelength</i>		

Light travels in the form of _____ and is part of the electromagnetic _____. All electromagnetic waves are a form of _____ and travel at approximately 300 000 kilometres per _____. They _____ travel through empty space (_____).

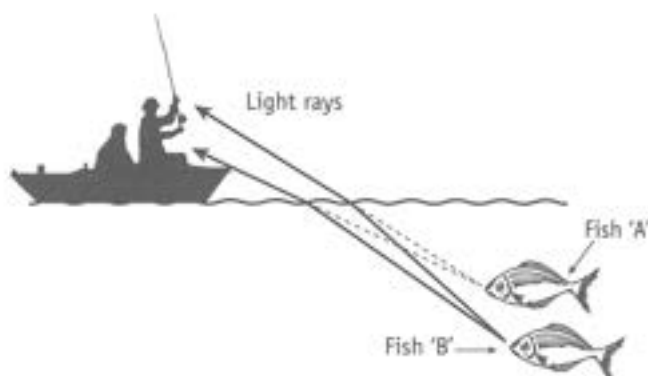
The diagram shows a _____ wave. The _____ of a transverse wave is measured by the distance between one crest (top) of a wave to the next. This is shown by 'X'. The _____ of a transverse wave is half the height of the wave and is shown by 'Y'. Other types of electromagnetic waves are _____ waves, microwaves, X-rays, and cosmic rays.



Light travels in straight lines but can be made to change _____ and diffract (_____) by going through different _____ materials. It travels at 300 000 km per second in air, 225 000 km per second in water and 200 000 km per second in glass. As a beam of light travelling in air enters a glass prism its speed _____. This is because glass is more _____ than air. When it goes from glass to air, its speed _____. It is then refracted in the opposite direction. The _____ is a line drawn at 90° to the surface of the prism at the point where the beam of light enters or leaves the prism. The angle shown by i° in the diagram is the angle of _____ and the angle shown by r° is the angle of _____.



Refraction must be considered when fishing. Light rays from the fish are bent as they move from _____ to air. The _____ position of the fish is shown as 'A' and its _____ position is shown as 'B'.



Looking at Light: Cloze

If a ray of white light is shone through a triangular _____ it can be split into a spectrum because the colours that make up white light are refracted by _____ amounts. This effect is known as _____.

Convex lens

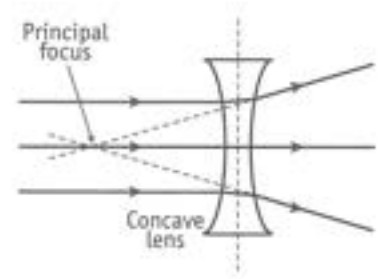
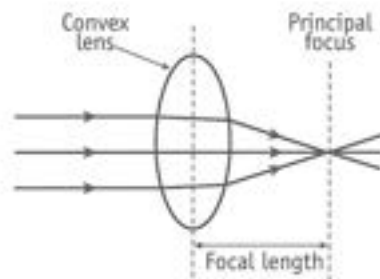


Lenses are made of transparent materials such as _____ or plastics. They are used to _____ (bend) light. _____ lenses have a surface that curves out and _____ lenses have a surface that curves in. A convex lens can be used as a magnifying glass — it gives an image which is _____ than the object. A concave lens gives an image which is _____ than the object.

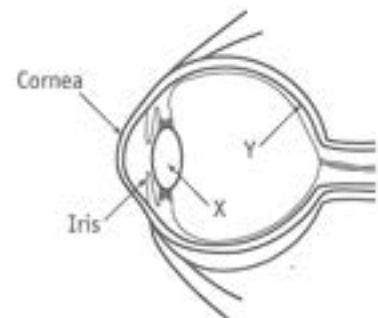
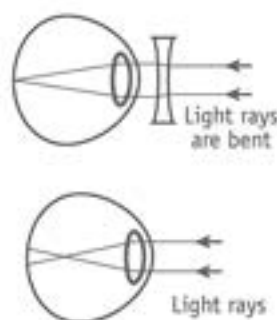
Concave lens



When parallel rays pass through a convex lens they _____ and come together at the principal _____. When parallel lines pass through a concave lens they _____. The principal focus is the point from which the light rays _____ to come.



The human _____ also has a convex lens. Light first comes into the eye through the transparent _____. The _____ opens and closes to change the amount of

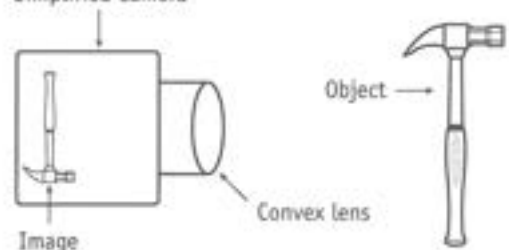


light entering the eye. The part labelled X in the diagram is the _____ which helps to focus light on the part labelled Y, the _____.

Sometimes an eye lens does not work properly. A _____-sighted person is unable to see objects clearly which are far away. The lens in the eye brings light to a focus just in _____ of the retina. A concave lens is needed to bring the image to focus on the retina. A _____-sighted person is unable to see objects clearly which are close. The lens in the eye brings light to a focus just _____ the retina. A convex lens is used to bring the image to focus on the retina.

A simple _____ has a lens that gives an _____ image on the film when the camera shutter is opened.

Simplified camera

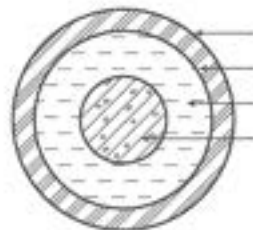


- 1 On the diagram, indicate the position of the film and the shutter.
- 2 On the diagram, show how rays of light travel from the object to form the image.

Inside the Earth

The Earth can be divided into four layers. The *crust* is the outer layer and it varies in thickness. It is believed to be about 5 kilometres thick under the oceans. Under the continents it is thought to vary from 15 km to 35 km in thickness. The *mantle* is the next layer and it is about 2900 km thick. This layer is believed to be semi-solid (like thick custard). The *outer core* is thought to be about 2200 km thick and made of liquid rock. The *inner core*, which has a radius of approximately 1300 km, is believed to be solid, probably iron (Fe) and nickel (Ni) under great pressure.

- 1 Copy the section diagram of the Earth on the right. Label the layers.
- 2 What is the approximate diameter of the Earth?
- 3 Which element has the symbol Ni?
- 4 Which element has the symbol Fe?



The Earth is not a solid ball. It is more like an orange. The crust is the thin skin holding both solid and liquid rock inside.

- 5 What is the approximate thickness of the crust under the seas?
- 6 What is the approximate thickness of the crust under the continents?



Within the Earth, it is very hot, and convection currents move through the liquid rocks. Hot molten rock, being less dense than cooler rock, rises to the surface. When it reaches the surface it cools and sinks again. These currents, together with the heat, mean that everything inside the Earth is under a lot of pressure. Since the Earth's crust is always moving, the features on the Earth are also changing all the time. The crust is constantly being pushed and pulled around by forces inside the Earth.

- 7 What does the science of geology study?

PLATE TECTONICS AND THE THEORY OF ISOSTASY

The Earth's crust is made up of seven main pieces, or *plates*, that float on the mantle, which is denser than the plates. The density of the Earth's crust decreases as the thickness increases. Rock materials under the Earth's crust flow and this causes some land masses to slowly rise while others fall. This theory is called *isostasy*. The plates are believed to be moving very slowly all the time. Those in the Pacific region are believed to be moving at about 4 cm per year.

Tectonics: a branch of geology that is involved with the nature of the Earth's crust.

- 8 What type of currents are causing the liquid rocks under the plates to move?

The major plates are the African, Eurasian, Indo-Australian, Pacific, North American, South American and Antarctic Plates. Major earthquakes are often associated with the edges of these plates.

- 9 Name a scale used to measure/describe the strength of earthquakes.
- 10 Name one other geological phenomenon which occurs where tectonic plates meet.



RESEARCH

- 11 Seismology is the study of which natural phenomena?
- 12 Give another definition for 'isostasy'.
- 13 What is the 'asthenosphere'?
- 14 What is the 'Moho (Mohorovicic) discontinuity'?

Inside the Earth

The German geologist Alfred Wegener argued that continental plates break, move apart, and can collide with each other. He noticed that the edges of some opposite continents matched. Also rock formations on opposite sides of the Atlantic, in South America and Africa, were very alike. They also contained fossils of similar animals. Tectonic activity (earthquakes and volcanic eruptions) occurs where plates meet. Plates move in different directions in different areas. There are four ways in which the edges of the plates move. Read the description given for each type and draw how the plates are moving. (Hint: decide whether you are looking at the plates from above or edge-on.)

- 15 *Sliding edge*: the edges of plates are moving across each other. Draw this.
- 16 *Subduction zone*: the edge of one plate goes under the other one. Draw this.
- 17 *Collision zone*: plates are moving towards each other. Draw this.
- 18 *Spreading/divergent zone*: plates are moving apart. Draw this.

Plates that just slide past each other without adding or taking away material are called *conservative plate boundaries*. A *transform fault* occurs where the fault plane is vertical and the plates touch each other. Most of these faults occur in oceans — however, one famous example is the San Andreas fault near San Francisco. The plates do not slide past each other smoothly due to friction.

- 19 What will happen if the plates suddenly slip?

Subduction is when one plate slides under another. One above-sea example is where the (European) Southern Alps meet and slide beneath the Northern Alps. If one plate is not pushed into a subduction zone, the two plates can be pushed upward together, forming mountain ranges.

Where plates are moving apart, they stretch and the edges become thinner. Magma (molten rock) pushes upward and is extruded between the plates. It cools down and forms new plate material. One example of this is the Mid-Atlantic ridge where the largest chain of submarine mountains exist. This ridge has a line of volcanoes along it. Some, such as Iceland, are tall enough to form islands. These types of boundaries are called *constructive plate boundaries*. When they occur on land they are called *rift valleys*.

- 20 Who first came up with the theory of *continental drift*?
- 21 What evidence did the scientist in Question 20 use to support his theory?
- 22 What is meant by a *submarine mountain range*?
- 23 Iceland was formed from a volcano. True or false?
- 24 Are constructive plate boundaries moving together or moving apart?

Ocean ridges are where new ocean crust is being created. Hot molten rock is carried to the surface by convection currents. It is cooled and solidified by sea water. The plates on either side of the ridge are moving apart at a speed varying from 1 cm to 2 cm a year. This slow movement has, over millions of years, brought about continental drift.

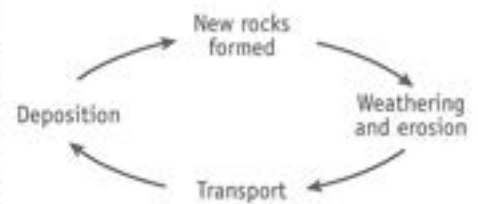
- 25 Give another name for 'hot molten rock' that comes from inside the Earth.
- 26 Explain why, if the ocean floor is being rifted apart in one part of the Earth, the resulting expansion must be balanced by contraction elsewhere.
- 27 Rocks near a mid-ocean rift will be younger than those far from the rift. True or false?

There are places (destructive plate boundaries) where plates are pushed together. One plate is pushed under the other. Rock is carried down into the mantle where it melts. This produces new magma which erupts on the crust. These types of eruptions have made chains of volcanic islands, such as Japan and the Philippines.



The Rock Cycle

The Earth's crust (*lithosphere*) is made of rocks and soil. The three main types of rock are *igneous*, *sedimentary* and *metamorphic*. They have different properties and appearances depending upon how they were formed. Molten (liquid) rock that has cooled to form solid rock is called igneous rock. Igneous rocks usually contain crystals. They can come from volcanoes (*volcanic rock*) or can be formed deep inside the Earth (*plutonic rock*). Basalt and granite are two types of igneous rock. The process by which rocks are formed, broken down, moved and reformed over and over again is called the *rock cycle*.



Every day various physical and chemical processes are at work changing the appearance of the Earth's surface. Rocks that cover the Earth's surface are being broken down into smaller pieces by wind, rain, ice, plants and temperature changes. For example, rain can be slightly acidic and can dissolve some kinds of rock. This slow breakdown is called *weathering*.

Weathering: the breakdown of rock into smaller fragments.

Erosion: the removal or transportation of rock fragments.

Once a surface has been weakened, it can be worn away. The removal or transportation of soil or pieces of rock is called *erosion*. Erosion results mainly from wind, running water or rain. Humans can cause erosion, for example, gully erosion, banks of rivers, clearing of land, soil tillage and so on. If too many animals are kept on an area of land, then overgrazing can occur. The grass and roots are eaten and the topsoil is not held together. This can then be blown away by strong winds or washed away by heavy rain.

Weathering and erosion, over a long period of time, can cause river valleys, caves, sand dunes, rock shapes and cliffs. Even hard rocks can eventually be broken down into very small particles, eventually producing material called *sediment*. Sediment is deposited into rivers, lakes and seas and eventually settles on the bottom. Layers of sediment are then compacted and compressed to form horizontal layers of sedimentary rock.

Over many years, layers of sedimentary material build up — the bottom layer is the oldest, the top is the youngest. It can take a long time for small pieces of rock to be eroded from a surface, transported to another place and then deposited. It can take millions of years for the small pieces to be compacted together and changed into hard rock. Common sedimentary rocks are sandstone, limestone, coal, conglomerate, shale and mudstone. Many sea creatures have shells made out of calcium carbonate (CaCO_3). When they die, their shells fall to the ocean floor and also form a sediment. They eventually form limestone. When igneous or sedimentary rocks are changed by great heat and pressure underground they are called *metamorphic* rocks. Most metamorphic rocks are layered. Slate and marble are metamorphic rocks. They have very different properties from the rocks from which they were formed.

- 1 What is the 'lithosphere'?
- 2 What do 'plutonic' and 'volcanic' rocks have in common?
- 3 To what group of rocks does basalt belong?
- 4 Define 'weathering'.
- 5 Name two things which cause weathering.
- 6 Define 'erosion'.
- 7 Name two things that cause erosion.
- 8 What is the general name for rocks which have been changed by heat and pressure?
- 9 Name two common sedimentary rocks.
- 10 What is the name and formula of the chemical that makes up limestone?
- 11 What do we call the process by which rocks are formed, broken down, moved and reformed over and over again?

Tectonic Plates: Assignment

NAME _____ DUE DATE _____

Use the map of the Earth below to label and show the borders/locations of the following features:

- | | | |
|----------------------------|----------------------------|-----------------------------|
| 1 The African Plate | 2 The Eurasian Plate | 3 The Australia-India Plate |
| 4 The Pacific Plate | 5 The North American Plate | 6 The Nazca Plate |
| 7 The South American Plate | 8 The Antarctic Plate | 9 The Mid-Atlantic Ridge |
| 10 The San Andreas Fault | 11 The Marianas Trench | |



Say whether each of the following statements is true or false.

- 12 Seismic waves can be used to study the interior of the Earth. _____
- 13 S waves can travel through solids but not liquids. _____
- 14 L waves are the fastest seismic waves and travel through the core. _____
- 15 Oceanic rock tends to be basaltic. _____



RESEARCH

- 16 Define 'geophysics'.
- 17 Discuss Victor Hugo Benioff's contributions to the study of geology.
- 18 What is a 'tsunami'?

Folds and Faults

Sediment: small particles that settle to the bottom of a liquid, the pieces are deposited by water, wind, or glaciers.

Sedimentary rocks are very common. They are made up of small bits of rock which have been squashed together. Sedimentary rocks, such as sandstone and limestone, tend to form layers or *strata* over long periods of time. These layers do not always stay horizontal. They are subjected to forces from all directions.

Rocks become folded when they are squeezed by compression forces. Mountains such as the Alps, Himalayas and the Rockies have been formed by folding over millions of years.

- 1 Name the location of the Alps.
- 2 Name the location of the Himalayas.
- 3 Name the location of the Rockies.

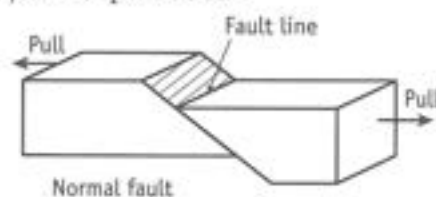


An upward fold of layers of rock is called an *anticline* and a downward fold is a *syncline*.

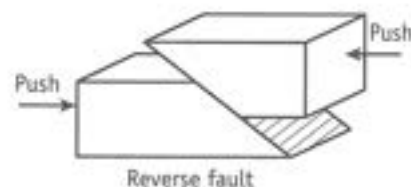
4 Copy the diagram on the right. 5 Identify part 5 on the diagram. 6 Identify part 6 on the diagram. Folds occur when rocks which are ductile are subjected to compression forces. Sometimes, rocks do not bend when compressed — the layers may break. When this happens, a *fault* is produced.

- 7 Define 'ductile'.
- 8 Define 'sediment'.
- 9 Name two types of sedimentary rock.
- 10 Why can the fossils of sea creatures sometimes be found on the top of fold mountains?

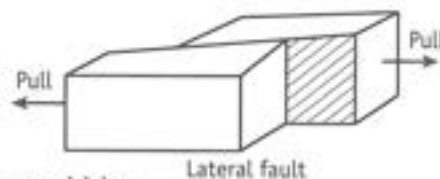
Faults occur in layers of rock when they are stretched or compressed past their breaking point. There are three main types of fault: *normal*, *reverse* and *lateral*. In a normal fault, the rock drops down on one side, sometimes by several metres. If this sliding continues, an escarpment (cliff) many hundreds of metres deep can form.



Normal fault



Reverse fault



Lateral fault

- 11 Copy the diagram of a normal fault and show where an escarpment would be.

A reverse fault is caused by compression forces. It affects the landscape in a similar manner to a normal fault, but one side of the fault slides over the other side.

- 12 Copy the diagram of a reverse fault and show where the fault line would be.

A lateral (transcurrent, wrench) fault occurs when the rock on one side of a fault line is moving sideways in the opposite direction to the rock on the other side. Escarpments may form along a lateral fault, although most movement is sideways.

- 13 Where do very large fault lines occur?

The San Andreas Fault (USA) and the Alpine Fault (NZ) are examples of lateral faults. If there are several parallel fault lines in an area, the ground between these faults can subside to form a rift valley, such as Death Valley (California, USA).

- 14 What is an 'escarpment'?
- 15 What geological events are associated with large lateral faults?

Answer true or false to each of the following.

- 16 Normal faults are caused by extensional forces. _____
- 17 Rocks need time, temperature and pressure to bend. _____
- 18 Reverse faults extend the Earth's crust. _____

Geological Periods

INSTRUCTIONS

Cut out the title 'Geological Periods' and place this at the top of your page. Cut out the 'Introduction' and put this below the title. Cut out the rest of the boxes and arrange them in a time-line, that is, in order of age. Put the oldest periods at the bottom, the newest at the top. Don't stick them in until you are sure of the order. (Note: different sources may give slightly different times for these periods.)

INTRODUCTION

There are few fossils of animals from earlier than about 600 million years ago. From around 570 million years ago the number of fossils increased. The time from then to now is divided into three *eras*, the *Paleozoic* (570 million to 225 million years ago), the *Mesozoic* (225 million to 65 million years ago) and the *Cainozoic* (65 million years ago to the present day). These eras are sub-divided into *periods*.

CAMBRIAN (570-500 million years ago.) Fossils of sea life in increasing numbers.

CARBONIFEROUS (345-280 million years ago.) Swamps covered in plants. First reptiles.

CRETACEOUS (135-65 million years ago.) Dinosaurs become extinct, first flowers.

DEVONIAN (395-345 million years ago.) First animals found on land — amphibians.

JURASSIC (190-135 million years ago.) Dinosaurs common, first birds appear.

ORDOVICIAN (500-440 million years ago.) First vertebrates (fishes) appear.

PERMIAN (280-225 million years ago.) Reptiles common, first conifers appear.

QUATERNARY (2 million years ago to today.) First people appear.

SILURIAN (440-395 million years ago.) First land plants.

TERTIARY (65-2 million years ago.) Mammals and flowers common.

TRIASSIC (225-190 million years ago.) First dinosaurs and mammals.



RESEARCH

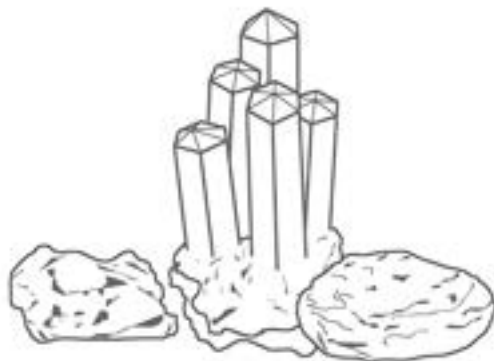
Find one additional fact about each period.

Rocks and Minerals: Questions

- 1 The Earth's crust is made up of many kinds of rock, but these can be split into three main groups. What are these groups?
- 2 'Igneous rocks were once molten rocks'. What is the name for molten rock deep within the Earth?
- 3 Igneous rocks contain crystals. Describe the features of crystals.

The type and size of crystals in igneous rocks depends on the rate at which the magma from which they were formed cooled.

- 4 What size of crystals are produced by rapid cooling?
- 5 What size of crystals are produced by slow cooling?
- 6 Basalt is a rock made of small crystals, formed from lava which has cooled quickly on the surface of the Earth. Granite has large crystals. Is it likely to have formed from lava? Explain your answer.
- 7 Are igneous rocks usually hard or soft.
- 8 What type of rocks are formed from grains of rock?
- 9 Apart from silt and sediment, what else can form sedimentary rock?
- 10 Chalk and limestone are both forms of what? (Hint: CaCO_3)



Sedimentary rocks tend to form in rivers, lakes and oceans. Often the remains of living creatures are buried in the sediment and can be preserved as fossils.

- 11 Are you likely to find fossils in igneous rocks? Explain your answer.

Copy and complete the following passage, replacing the numbers with words.

If sedimentary rock is subjected to (12) temperatures and pressures over a long period of time it can change to form (13) rock. Mudstone can be changed to (14), a black layered rock, chalk can be changed to (15) and sandstone can be changed to quartzite.

- 16 Are metamorphic rocks harder or softer than sedimentary rocks?
- 17 Some rocks have been formed from fragments of other rocks. Give one example.

Obsidian is a dark (often black) glass-like rock. It can be shaped easily to give sharp edges and was used by early peoples to make weapons and tools.

- 18 What type of rock is obsidian likely to be?

Rocks can be tested and identified by:

- their colour
 - their hardness
 - the presence of crystals
 - their density
 - how they react with acid.
- 19 Name one other property of rocks that can be used to help identify them.
 - 20 Study the table on the right. What property does it measure?
 - 21 What scale, named after the German mineralogist, does the table show?

Value	Property
1	Crushed by finger pressure.
2	Scratched by a coin.
3	Scratched by iron nail.
4	Scratched by glass.
5	Scratched by steel blade.
6	Scratched by quartz.
7	Will scratch glass.
8	
9	
10	As hard as diamond.

Rocks and Minerals: Cloze

Use the following words to fill in the spaces in the text below:

<i>acidic</i>	<i>appearances</i>	<i>basalt</i>	<i>coal</i>	<i>contour</i>	<i>crystals</i>	<i>cycle</i>
<i>diamond</i>	<i>erosion</i>	<i>geologist</i>	<i>glaciers</i>	<i>heat</i>	<i>humans</i>	<i>igneous</i>
<i>lava</i>	<i>limestone</i>	<i>magma</i>	<i>marble</i>	<i>metamorphic</i>	<i>millions</i>	<i>mineral</i>
<i>ore</i>	<i>plutonic</i>	<i>quartz</i>	<i>sandstone</i>	<i>sediment</i>	<i>wind</i>	<i>weathering</i>

A _____ is a rock which contains substances which are useful to us. Often these are in regular shapes called _____. Some common minerals are _____, feldspar, mica and haematite. A mineral which contains enough of a metal to make it worth extracting from the ground is an _____. The three main types of rock are sedimentary, igneous and _____. They have different properties and _____ depending upon how they were formed. A person who studies rocks and minerals is called a _____.

Molten (liquid) rock that has cooled to form solid rock is called _____ rock. Igneous rocks usually contain crystals. They can come from volcanoes (*volcanic rock*) or can be formed deep inside the Earth (_____). _____ and granite are two types of igneous rock. The molten rock deep inside a volcano is called _____ and the molten rock coming out of a volcano is called _____.

All rocks are eventually broken down into small pieces by the weather. When these pieces become small enough, they are moved by water, _____ or ice and usually end up in the ocean. This _____ forms layers and is gradually compressed (squashed) to form hard rock. The most common sedimentary rock is _____.

Many sea creatures have shells made out of calcium carbonate. When they die, their shells fall to the ocean floor and also form a sediment, which eventually forms _____. Other sedimentary rocks include shale, conglomerate and _____ (an impure form of carbon).

When igneous or sedimentary rocks are changed by great _____ and pressure underground they are called *metamorphic* rocks. Many metamorphic rocks are layered. Slate (black) and _____ (white) are metamorphic rocks. They have very different properties from the rocks from which they were formed. Carbon can form _____ (the hardest known naturally occurring substance) under certain conditions.

The process by which rocks are formed, broken down, moved and reformed over and over again is called the rock _____. It can take a long time for small pieces of rock to be eroded from a surface, transported to another place and then deposited. It can take _____ of years for the small pieces to be compacted together and changed into hard rock. Outdoors, the surface of a material can be weakened by the action of the weather. This is called _____. For example, rain can be slightly _____ and can dissolve some kinds of rock. Once a surface has been weakened, it can be worn away. This is called _____. Running water, the sea, the wind and _____ (ice) can cause erosion. _____ can also cause erosion, for example, gully erosion, banks of rivers, clearing of land, soil tillage and so on. We must guard against this and take measures to minimise the effects, by _____, ploughing and drill seeding for example.

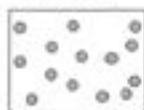
A1 AND 2 AN ELEMENTAL TALE

Li is lithium, Ce is cerium, Au is gold, Hg is mercury, P is phosphorus, Ca is calcium, Zn is zinc, S is sulfur, Sb is antimony, I is iodine, In is indium, Kr is krypton, Cl is chlorine, Eu is europium, B is boron, Ge is germanium, Fe is iron, Bi is bismuth, Rh is rhodium, Ne is neon, U is uranium, Br is bromine, Ba is barium, Ra is radium, Pd is palladium, Ta is tantalum, Sn is tin, Pb is lead, Mg is magnesium, Cu is copper, Ni is nickel, Pt is platinum, Ag is silver, Xe is xenon, Si is silicon, H is hydrogen, Cr is chrome, C is carbon, Ar is argon.

A3 AND 4 MATTER

- 1 Solid: particles are close together, and in regular order. Liquid: particles are close together, in no regular order. Gas: particles are far apart, in no regular order.

2



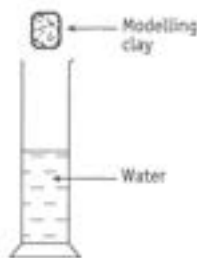
Air particles



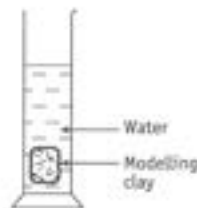
Air particles after compression

- 3 No, the particles are already close together.
4 Show solid particles vibrating and rotating about a fixed position.
5 Show liquid particles moving in a random manner.
6 Show gas particles moving in a random manner but greater than liquid particles, and further apart.
7 Yes, there are air spaces in and between the granules.

8



Water



Water

Modelling clay

- 10 No, it is the same amount of solid.
11 Melting.
12 Boiling (vaporisation).
13 Evaporation.
14 Particles in crystal are leaving the solid and move until they are evenly spaced throughout the liquid.
15 Liquid perfume evaporates and changes to gas particles. These move and eventually go up your nose.
16 Moving.
17 Matter is any physical substance or material. It is anything that takes up space.
R18 The change between the solid and gaseous states without the intermediate liquid state.
R19 Changing from a gas (vapour) to a liquid.
R20 Random motion of small particles being hit by smaller particles, such as smoke and air.

A5 AND 6 CHEMICAL REACTIONS

- 1 H
2 O
3 Oxygen.
4 Iron oxide.
5 Two to three.
6 Physical.
7 Is not.
8 Reaction.
9 Always.
10 Oxygen, fuel and heat.
11 The fire would go out.
12 Chemical reactions are usually difficult to reverse.
13 Physical: water boiling, freezing an 'icy pole', clothes drying. Chemical: making toast, candle burning, cooking a sausage, apple rotting, gas burning.
14 Iron (Fe), oxygen (O).
15 Tungsten (W).
16 Two atoms of hydrogen and one atom of oxygen are present in each molecule of water.
17 An explosion.
18 Corrosion (rusting).
19 Rusting.
20 Reactants are the materials you start with and products are the new substances formed.
21 Photosynthesis, digestion, etc.
22 Tooth decay, corrosion, etc.
23 Glucose.
24 Exothermic — heat is given out.
R25 Paint: coat with oil/grease; plate in another metal, such as zinc, tin or chrome; keep dry; etc.

A7 AND 8 SEPARATION TECHNIQUES

- 1 75%
2 No, the water has substances dissolved in it.
3 Atmosphere.
4 Mixtures: sea water, soda water, 18 karat gold, air, copper sulfate solution, blood, milk, petrol, soil and steel. Pure: distilled water, water, copper sulfate, lead, silver, oxygen, carbon dioxide, helium, iron and sugar.
5 It has pores (tiny holes).
6 No, both dissolve in water.
8 Vapour.
9 Water out.
10 Liebig condenser.
11 Distillate.
12 Cold water in.
13 Heat.
14 Solution.
15 They can mix together.
16 Place the container under the funnel. Open the tap and allow the water to flow into the container. Close the tap when the oil reaches tap. Replace the container. Open tap and let the oil flow.
17 Oil, it floats on top of water.
18 $\text{NH}_4(\text{s}) \rightleftharpoons \text{NH}_4(\text{g})$
19 Ammonium chloride will turn to a gas, leaving sodium chloride behind.
20 Spin drier.
21 No, salt is dissolved.
22 Colour would be less intense.
23 Evaporate water.
24 Pour off upper liquid layer, leaving solid behind.
25 Upper layer (usually clear) above settled material.
26 One that can dissolve no more solute.
27 When a solid changes directly to a gas (and vice versa) without changing to a liquid.

B1 NEWTON'S LAWS OF MOTION: LAW I

- 1 For example, 'Without any forces acting on it, an object that is moving, will continue to move at constant velocity'.
2 Gravity.
3 Pushing up with equal force.
4 The craft will continue to move at constant velocity, there is no friction or gravity to affect it.
5 Air pushes the puck above the table, so friction is much less.
6 Yes. It will continue to travel at the original velocity unless a force acts on it.
7 True.
8 When the car stops suddenly, they can be flung into the front of car, causing damage/injury. See Newton's 3rd law.
R9 The study of forces and motion.
R10 He lived 1642–1727, in England.
R11 The laws of falling bodies, projectile motion, the pendulum, etc.
R12 He invented the reflecting telescope, did work on gravity and the planets, etc.

B2 NEWTON'S LAWS OF MOTION: LAW II

- 1 Four times the acceleration, that is 4g
2 Very small, large number of newtons involved in only a moderate force.
3 $2000 \times 3 = 6000 \text{ N}$
4 Car B, it has less mass to accelerate with the same force.
5 No, a force is needed before movement can take place.
6 Both have the same inertia as they have the same mass.

B3 NEWTON'S LAWS OF MOTION: III

- 1 There is no air in space.
2 When you hit the nail with the hammer (action) there is an equal and opposite force (reaction) from the nail. The blunt end pushes in the opposite direction to the motion of the hammer, while the pointed end of the nail goes into the wood.
3 The racquet exerts a force on the ball in the opposite direction to which the ball is travelling. This force is greater than that exerted by the ball and causes it to move in the direction of the racquet. The racquet will slow down on impact with the ball.
4 It will be squashed inwards where it is in contact.
5 On stepping onto the jetty the fisherman would push on the boat. This force moves the boat.
6 The flask rotates. This is because the steam exerts a force (similar to a rocket) as it comes out of the tubes.
7 False.
R8 Approx 20 to 62. Alexandria (Egypt), or possibly Greece.

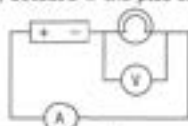
B4 SIMPLE MACHINES I

- 1 A substance which lessens friction and makes things 'slippery'.
2 They couldn't grip on the road, they would slip.
3 Pushes and pulls (also twists and turns).
4 Choose from gravity, magnetism, electricity, nuclear.
5 The newton.
6 Load, fulcrum (pivot) and effort.

C5 OUTPUT OF A POWER SUPPLY

1 DC, because of the plus and minus signs shown.

2



3 The brightness of the lamp would vary.

4 Ammeter.

5 Voltmeter.

6 $\Omega = V \div A$ 7 $W = V \times A$

8

Power (W)	Resistance (Ω)
0.98	2
2.61	3.2
4.84	4.0
8.26	4.2
11.8	4.89
15.8	4.89
21.6	4.9

9 No, the light stays on, the circuit is not broken (the voltmeter is wired in parallel).

10 Yes, the light goes out, the circuit is broken (the ammeter is wired in series).

11 12 V, 24 W to cope with highest voltage (10.3 V) and power (21.6 W).

12 Electrons.

R13 A device which changes AC to DC.

C6 FUSES AND CURRENT1 Ohms, Ω .

2 The tube is an electrical insulator, it offers resistance to heat.

4 Diagram A.

5 Metal.

6 It is a conductor.

9 Switch.

10 Fuse.

11 Active/live wire.

12 Earth/ground.

13 Neutral wire.

14 Resistor/heating element.

R15 They are devices which switch off when an electrical fault occurs. They can be reset.

R16 It could heat up, the insulation could melt and cause a fire.

Amps (A)
0.25
1.5
3.13
2.29
8.33

C7 TRANSFORMERS

1 Alternating current.

2 It reverses direction (polarity) 50 times a second.

3

Primary turns	Primary voltage	Secondary turns	Secondary voltage
100	12	1000	120
20	12	100	60
50	240	25	12
1833	110	200	12
300	240	2100	1680

4 Electricity output is 99% of electrical input, that is 1% is 'lost' converting the energy.

5 So they don't short circuit.

6 Secondary.

7 Less.

8 It is safer.

R9 Substations use step-down transformers to drop the voltage, from 110 000 V for example.

R10 Tesla: created artificial lightning, and a wireless communications system, discovered the principle of rotating magnetic field, etc. Oersted: first person to isolate aluminium, discovered that a magnetic needle is deflected by a current-carrying wire, etc.

R11 No, DC (direct current) doesn't change direction, so it can't induce current in a secondary coil.

C8 ELECTRICAL TERMS

1 t

2 c

3 x

4 h

5 w

6 a

7 m

8 n

9 u

10 d

11 o

12 q

13 e

14 l

15 v

16 i

17 p

18 r

19 b

20 f

21 j

22 s

23 g

24 k

25 z

26 y

6 Doppler Effect — higher frequency when an object is moving towards you, lower frequency when it is moving away.

7 $2 + 3 = 0.67$ s.

R8 They fly faster than the speed of sound (Mach 1).

R9 For example, when an object vibrates at its natural frequency, when a sound of the same frequency occurs nearby.

D2 ULTRASOUND

1 20 kHz.

2 They send out ultrasonic waves and listen to echoes.

3 'Silent' dog whistles, sonar, scanning the human body, cleaning jewellery, etc.

4 The range of frequencies we can hear becomes less.

5 Because body parts reflect ultrasound differently.

6 Considered to be less harmful than X-rays.

7 Fish finding, to find the depth of the water.

8 To determine the depth of water, to detect underwater obstacles.

9 To detect submarines.

10 So that only the dog will hear it.

R11 Supersonic: faster than the speed of sound, Ultrasonic: a sound with frequency too high to be heard.

R12 They emit clicks and whistles. Clicks are for echo location. Whistles are for communication, etc.

D3 HEARING LOSS

2 34 dB to 35 dB.

3 33 years.

4 Hearing loss continues to worsen at a regular rate and that they had suffered no extra hearing damage.

5 13 dB (compared to a 25 year old).

6 Two times.

7 Eight times.

R8 Noisy workplace, loud rock music, inherited hearing loss, ageing, etc.

R9 Disease (for example, otitis media) and accidents are likely to be more common.

R10 They can damage the hair cells of the cochlea, etc.

R12 They are devices which change sound to electrical impulses relayed directly to the cochlea, stimulating the auditory nerve.

D4 NOISE AND HEALTH

1 A loud, unpleasant/unmusical sound.

2 decibels (dB).

3 Traffic, airports, public transport, building work and machinery, etc.

4 Prolonged exposure to loud noise.

5 Tiredness.

6 Feeling sick.

7 When working it will generate maximum noise level of 77 decibels.

8 Double glazing, heavier curtains, screening by trees, a high fence, etc.

9 The quietest sound you can hear.

10 Ear protection.

11 Playing them too loud for too long can damage hearing.

R14 It was designed in such a way that the sound quality was poor.

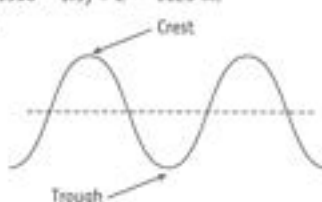
D5 SOUND: EXTENDED QUESTIONS

1 Deep water, it has further to travel.

2 1600 m (800 m down and 800 m back up).

3 330 m.s^{-1} 4 $1600 \div 1500 = 1.1$ seconds.5 $(1500 \times 1.5) \div 2 = 1125 \text{ m}$.

7 and 8



9 Wavelength.

10 Amplitude.

11 Quickly.

12 Gases.

13 Longitudinal (compression) wave.

14 Sound.

15 $330 \div 5000 = 0.066 \text{ m}$.

16 The Doppler Effect.

17 A shift in frequency is caused by the movement of the sound-generating object relative to the listener.

18 decibels (dB).

D6 RADIATION

1 Ultra-violet.

2 Infra red.

3 The Sun and other celestial objects.

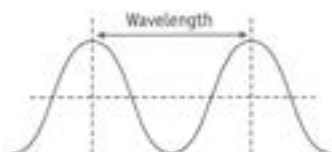
4 $300\,000 \text{ km.s}^{-1}$

5 Radioactive isotopes — they have large, unstable atoms.

6 IR, UV, visible light and radio waves.

7 The number of cycles per second passing a certain point.

8



A 'wavelength' is from crest (peak) to crest or trough to trough.

9 Hertz, Hz.

10 Metres, m.

11 They can pass through the body and be used to take pictures.

12 Short-wavelength.

R13 They can be used to generate X-rays for photographs, can be placed into a patient and traced through the body, etc.

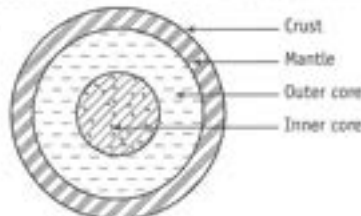
R14 A space that contains absolutely no matter.

D7 AND 8 LOOKING AT LIGHT

Waves, spectrum, energy, second, can, vacuum, transverse, wavelength, amplitude, radio, speed, bend, transparent, decreases, dense, increases, normal, incidence, reflection, water, apparent, real, prism, different, dispersion, glass, refract, convex, concave, larger, smaller, converge, focus, diverge, seem, eye, cornea, iris, lens, retina, short, front, long, behind, camera, inverted.

E1 AND 2 INSIDE THE EARTH

1



2 6400 km (approximately).

3 Iron.

4 15 km to 35 km.

5 Convection currents.

6 Volcanoes, mountain ranges, rifts, etc.

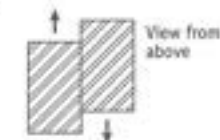
R11 Earthquakes.

R12 The equilibrium in the Earth's crust caused by subsurface rock flowing under stress.

R13 The area beneath the lithosphere which can move or flow.

R14 The boundary between the crust and mantle which has an approximate depth of 5 kilometres beneath ocean floors to about 40 kilometres under continents.

15



16



17



18



19 An earthquake.

20 Alfred Wegener.

21 The edges of opposite continents matched and had similar fossils.

22 Undersea mountains ('sub' means below, 'marine' means waters).

23 True.

24 Together.

25 Magma (inside the Earth) or lava (on the surface).

26 Isostasy, equilibrium, the land rising must be balanced by the same amount falling, etc.

27 True.

E3 THE ROCK CYCLE

1 Anything forming the solid part of the Earth's crust.

2 Both were once molten.

3 Igneous.

4 The slow breakdown of material.

5 Heat, cold, wind, rain, etc.

6 The removal of material once it has been broken down.

7 Wind, water, humans, etc.

8 Metamorphic.

9 Sandstone, limestone, coal, conglomerate, shale and mudstone.

10 Calcium carbonate, CaCO_3 .

11 The rock cycle.

E4 TECTONIC PLATES

1-11 Teacher to check student response.

12 True.

13 True.

14 False.

15 True.

R16 A branch of science applying physical principles to the study of the Earth.

R17 For example: work in seismology, mountain formation, Earth's magnetism, etc.

R18 A large sea wave produced by an earthquake or volcanic activity.

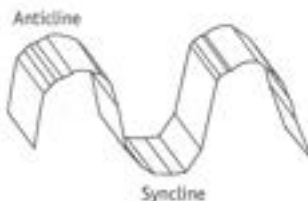
E5 FOLDS AND FAULTS

1 South, central Europe (Switzerland, Italy, France, Austria).

2 South Asia (India, Pakistan, Tibet, Nepal).

3 North America (Alaska to New Mexico).

4, 5, 6



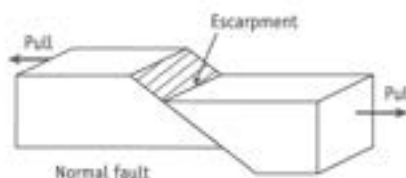
7 Can be drawn out.

8 Small pieces which settle, for example, to the bottom of a liquid.

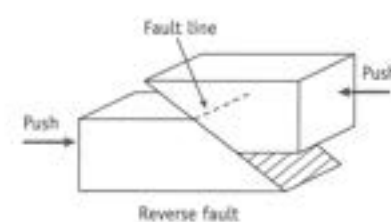
9 Sandstone, limestone, etc.

10 The earth has moved, for instance by folding, and material has been carried up to what becomes the top of the mountain.

11



12



13 Where tectonic plates meet.

14 A cliff.

15 Earthquakes and volcanoes.

16 True.

17 True.

18 False.

E6 GEOLOGICAL PERIODS

(Oldest to newest): Cambrian, Ordovician, Silurian, Devonian, Carboniferous, Permian, Triassic, Jurassic, Cretaceous, Tertiary, Quaternary.

R Teacher to check student response.

E7 ROCKS AND MINERALS: QUESTIONS

1 Igneous, metamorphic, sedimentary.

2 Magma.

3 Flat faces, regular repeating angles.

4 Small.

5 Large.

6 No, lava cools (on surface) too quickly to produce large crystals.

7 Hard.

8 Sedimentary.

9 Shells/skeletons of living creatures.

10 Calcium carbonate.

11 No, any fossils would have been destroyed by heat.

12 High.

13 Metamorphic.

14 Slate.

15 Marble.

16 Harder.

17 Conglomerate.

18 Igneous (volcanic).

19 Size of crystals, lustre, etc.

20 Hardness.

21 Mohs' Scale.

E8 ROCKS AND MINERALS: CLOZE

minerals, crystals, quartz, ore, metamorphic, appearances, geologist, igneous, plutonic, basalt, magma, lava, wind, sediment, sandstone, limestone, coal, heat, marble, diamond, cycle, millions, weathering, acidic, erosion, glaciers, humans, contour.



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